

The meaning of this is that if state s occurs then consumer i has endowment vector $(\omega_{1si}, \dots, \omega_{Lsi}) \in \mathbb{R}^L$.

The preferences of consumer i may also depend on the state of the world (e.g., the consumer's enjoyment of wine may well depend on the state of his health). We represent this dependence formally by defining the consumer's preferences over contingent commodity vectors. That is, we let the preferences of consumer i be specified by a rational preference relation $\succeq_i$ defined on a consumption set $X_i \subset \mathbb{R}^{LS}$.

Example 19.B.1: As in Section 6.E, the consumer evaluates contingent commodity vectors by first assigning to state s a probability π_{si} (which could have an objective or a subjective character), then evaluating the physical commodity vectors at state s according to a Bernoulli state-dependent utility function $u_{si}(x_{1si}, \dots, x_{Lsi})$, and finally computing the expected utility.² That is, the preferences of consumer i over two contingent commodity vectors $x_i, x'_i \in X_i \subset \mathbb{R}^{LS}$ satisfy

$$x_i \succeq_i x'_i \quad \text{if and only if} \quad \sum_s \pi_{si} u_{si}(x_{1si}, \dots, x_{Lsi}) \geq \sum_s \pi_{si} u_{si}(x'_{1si}, \dots, x'_{Lsi}).$$

■

It should be emphasized that the preferences $\succeq_i$ are in the nature of *ex ante* preferences: the random variables describing possible consumptions are evaluated before the resolution of uncertainty.

Similarly, the technological possibilities of firm j are represented by a production set $Y_j \subset \mathbb{R}^{LS}$. The interpretation is that a (*state-*)*contingent production plan* $y_j \in \mathbb{R}^{LS}$ is a member of Y_j if for every s the input-output vector $(y_{1sj}, \dots, y_{Lsj})$ of physical commodities is feasible for firm j when state s occurs.

Example 19.B.2: Suppose there are two states, s_1 and s_2 , representing good and bad weather. There are two physical commodities: seeds ($\ell = 1$) and crops ($\ell = 2$). In this case, the elements of Y_j are four-dimensional vectors. Assume that seeds must be planted before the resolution of the uncertainty about the weather and that a unit of seeds produces a unit of crops if and only if the weather is good. Then

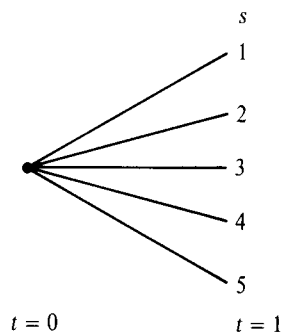
$$y_j = (y_{11j}, y_{21j}, y_{12j}, y_{22j}) = (-1, 1, -1, 0)$$

is a feasible plan. Note that since the weather is unknown when the seeds are planted, the plan $(-1, 1, 0, 0)$ is not feasible: the seeds, if planted, are planted in both states. Thus, in this manner we can imbed into the structure of Y_j constraints on production related to the timing of the resolution of uncertainty.³ ■

To complete the description of an economy in a manner parallel to Chapters 16 and 17 it only remains to specify ownership shares for every consumer i and firm j . In principle, these shares could also be state-contingent. It will be simpler, however, to let $\theta_{ji} \geq 0$ be the share of firm j owned by consumer i whatever the state. Of course $\sum_j \theta_{ji} = 1$ for every i .

2. The discussion in Section 6.E was for $L = 1$. It extends straightforwardly to the current case of $L \geq 1$.

3. A similar point could be made on the consumption side. If, for a particular commodity ℓ , any vector $x_i \in X_i$ is such that all entries $x_{\ell si}$, $s = 1, \dots, S$, are equal, then we can interpret this as asserting that the consumption of ℓ takes place before the resolution of uncertainty.

**Figure 19.B.1**

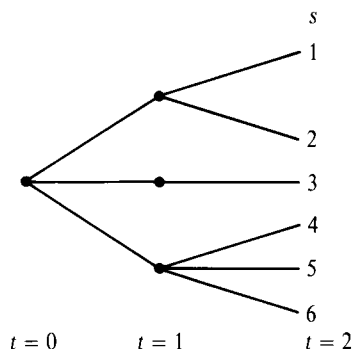
Two periods. Perfect information at $t = 1$.

Information and the Resolution of Uncertainty

In the setting just described, time plays no explicit formal role. In reality, however, states of the world unfold over time. Figure 19.B.1 captures the simplest example. In the figure, we have a period 0 in which there is no information whatsoever on the true state of the world and a period 1 in which this information has been completely revealed.

We have already seen (Example 19.B.2) how, by conveniently defining consumption and production sets, we can accommodate within our setup the temporal structure of Figure 19.B.1: a commodity that has as part of its physical description its availability at $t = 0$ should never appear in differing amounts across states.

The same methodology can be used to incorporate into the formalism a much more general temporal structure. Suppose we have $T + 1$ dates $t = 0, 1, \dots, T$ and, as before, S states, but assume that the states emerge gradually through a *tree*, as in Figure 19.B.2. These trees are

**Figure 19.B.2**

An information tree: gradual release of information.

similar to those described in Chapter 7. Here final nodes stand for the possible states realized by time $t = T$, that is, for complete histories of the uncertain environment. When the path through the tree coincides for two states, s and s' , up to time t , this means that in all periods up to and including period t , s and s' cannot be distinguished.

Subsets of S are called *events*. A collection of events $\mathcal{S}$ is an *information structure* if it is a partition, that is, if for every state s there is $E \in \mathcal{S}$ with $s \in E$ and for any two $E, E' \in \mathcal{S}$, $E \neq E'$, we have $E \cap E' = \emptyset$. The interpretation is that if s and s' belong to the same event in $\mathcal{S}$ then s and s' cannot be distinguished in the information structure $\mathcal{S}$.

To capture formally a situation with sequential revelation of information we look at a *family* of information structures: $(\mathcal{S}_0, \dots, \mathcal{S}_t, \dots, \mathcal{S}_T)$. The process of information revelation makes the $\mathcal{S}_t$ increasingly fine: once one has information sufficient to distinguish between two states, the information is not forgotten.

Example 19.B.3: Consider the tree in Figure 19.B.2. We have

$$\begin{aligned}\mathcal{S}_0 &= (\{1, 2, 3, 4, 5, 6\}), \\ \mathcal{S}_1 &= (\{1, 2\}, \{3\}, \{4, 5, 6\}), \\ \mathcal{S}_2 &= (\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}). \blacksquare\end{aligned}$$

The partitions could in principle be different across individuals. However, except in the last section of this chapter (Section 19.H), we shall assume that the information structure is the same for all consumers.

A pair (t, E) where t is a date and $E \in \mathcal{S}_t$ is called a *date-event*. Date-events are associated with the nodes of the tree. Each date-event except the first has a *unique predecessor*, and each date-event not at the end of the tree has one or more *successors*.

With this temporal modeling it is now necessary to be explicit about the time at which a physical commodity is available. Suppose there is a number H of basic physical commodities (bread, leisure, etc.). We will use the double index ht to indicate the time at which a commodity h is produced, appears as endowment, or is available for consumption. Then x_{hts} stands for an amount of the physical commodity h available at time t along the path of state s .

Fortunately, this multiperiod model can be formally reduced to the timeless structure introduced above. To see this, we define a new set of $L = H(T + 1)$ physical commodities, each of them being one of these double-indexed (i.e., ht) commodities. We then say that a vector $z \in \mathbb{R}^{LS}$ is *measurable* with respect to the family of information partitions $(\mathcal{S}_0, \dots, \mathcal{S}_T)$ if, for every hts and hts' , we have that $z_{hts} = z_{hts'}$ whenever s, s' belong to the same element of the partition $\mathcal{S}_t$. That is, whenever s and s' cannot be distinguished at time t , the amounts assigned to the two states cannot be different. Finally, we impose on endowments $\omega_i \in \mathbb{R}^{LS}$, consumption sets $X_i \subset \mathbb{R}^{LS}$ and production sets $Y_j \subset \mathbb{R}^{LS}$ the restriction that all their elements be measurable with respect to the family of information partitions. With this, we have reduced the multiperiod structure to our original formulation.

19.C Arrow-Debreu Equilibrium

We have seen in Section 19.B how an economy where uncertainty matters can be described by means of a set of states of the world S , a consumption set $X_i \subset \mathbb{R}^{LS}$, an endowment vector $\omega_i \in \mathbb{R}^{LS}$, and a preference relation $\succeq_i$ on X_i for every consumer i , together with a production set $Y_j \subset \mathbb{R}^{LS}$ and profit shares $(\theta_{j1}, \dots, \theta_{jI})$ for every firm j .

We now go a step further and make a strong assumption. Namely, we postulate the existence of a market for every contingent commodity ℓs . These markets open *before* the resolution of uncertainty, at date 0 we could say. The price of the commodity is denoted $p_{\ell s}$. What is being purchased (or sold) in the market for the contingent commodity ℓs is commitments to receive (or to deliver) amounts of the physical good ℓ if, and when, state of the world s occurs. Observe that although deliveries are contingent, the payments are not. Notice also that for this market to be well defined it is indispensable that all economic agents be able to recognize the occurrence of s . That is, information should be *symmetric* across economic agents. This informational issue will be discussed further in Section 19.H.

Formally, the market economy just described is nothing but a particular case of the economies we have studied in previous chapters. We can, therefore, apply to our market economy the concept of Walrasian equilibrium and, with it, all the theory

developed so far. When dealing with contingent commodities it is customary to call the Walrasian equilibrium an *Arrow–Debreu equilibrium*.⁴

Definition 19.C.1: An allocation

$$(x_1^*, \dots, x_I^*, y_1^*, \dots, y_J^*) \in X_1 \times \dots \times X_I \times Y_1 \times \dots \times Y_J \subset \mathbb{R}^{LS(I+J)}$$

and a system of prices for the contingent commodities $p = (p_{11}, \dots, p_{LS}) \in \mathbb{R}^{LS}$ constitute an *Arrow–Debreu equilibrium* if:

- (i) For every j , y_j^* satisfies $p \cdot y_j^* \geq p \cdot y_j$ for all $y_j \in Y_j$.
- (ii) For every i , x_i^* is maximal for $\succeq_i$ in the budget set

$$\{x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*\}.$$

- (iii) $\sum_i x_i^* = \sum_j y_j^* + \sum_i \omega_i$.

The welfare and positive theorems of Chapters 16 and 17 apply without modification to the Arrow–Debreu equilibrium. Recall from Chapter 6, especially Sections 6.C and 6.E, that, in the present context, the convexity assumption takes on an interpretation in terms of risk aversion. For example, in the expected utility setting of Example 19.B.1, the preference relation $\succeq_i$ is convex if the Bernoulli utilities $u_{si}(x_{si})$ are concave (see Exercise 19.C.1).

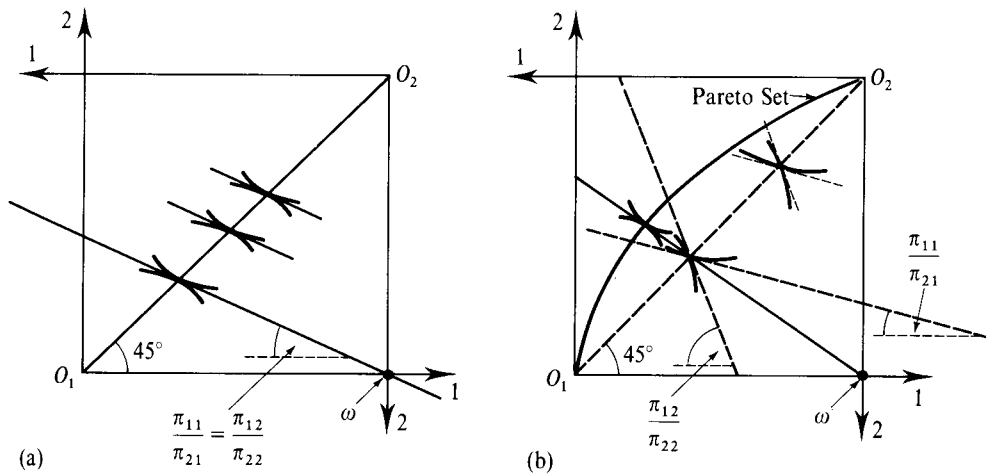
The Pareto optimality implication of Arrow–Debreu equilibrium says, effectively, that the possibility of trading in contingent commodities leads, at equilibrium, to an efficient allocation of risk.

It is important to realize that at any production plan the profit of a firm, $p \cdot y_j$, is a nonrandom amount of dollars. Productions and deliveries of goods do, of course, depend on the state of the world, but the firm is active in all the contingent markets and manages, so to speak, to insure completely. This has important implications for the justification of profit maximization as the objective of the firm. We will discuss this point further in Section 19.G.

Example 19.C.1: Consider an exchange economy with $I = 2$, $L = 1$, and $S = 2$. This lends itself to an Edgeworth box representation because there are precisely two contingent commodities. In Figures 19.C.1(a) and 19.C.1(b) we have $\omega_1 = (1, 0)$, $\omega_2 = (0, 1)$, and utility functions of the form $\pi_{1i}u_i(x_{1i}) + \pi_{2i}u_i(x_{2i})$, where (π_{1i}, π_{2i}) are the subjective probabilities of consumer i for the two states. Since $\omega_1 + \omega_2 = (1, 1)$ there is no aggregate uncertainty, and the state of the world determines only which consumer receives the endowment of the consumption good. Recall from Section 6.E (especially the discussion preceding Example 6.E.1) that for this model [in which the $u_i(\cdot)$ do not depend on s], the marginal rate of substitution of consumer i at any point where the consumption is the same in the two states equals the probability ratio π_{1i}/π_{2i} .

In Figure 19.C.1(a) the subjective probabilities are the same for the two consumers (i.e., $\pi_{11} = \pi_{12}$) and therefore the Pareto set coincides with the diagonal of the box (the box is a square and so the diagonal coincides with the 45-degree line, where the marginal rates of substitution for the two consumers are equal: $\pi_{11}/\pi_{21} = \pi_{12}/\pi_{22}$). Hence, at equilibrium, the two consumers insure completely; that is, consumer i 's equilibrium consumption does not vary across the two states. In Figure 19.C.1(b)

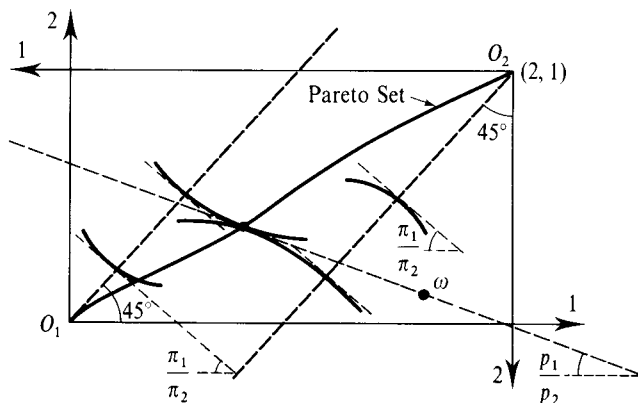
4. See Chapter 7 of Debreu (1959) for a succinct development of these ideas.

**Figure 19.C.1**

(a) No aggregate risk: same probability assessments.
 (b) No aggregate risk: different probability assessments.

the consumer's subjective probabilities are different. In particular, $\pi_{11} < \pi_{12}$ (i.e., the second consumer gives more probability to state 1). In this case, each consumer's equilibrium consumption is higher in the state he thinks comparatively more likely (relative to the beliefs of the other consumer). ■

Example 19.C.2: The basic framework is as in Example 19.G.1. The difference is that now there is aggregate risk: $\omega_1 + \omega_2 = (2, 1)$. The utilities are state independent and the probability assessments are the same for the two traders: (π_1, π_2) . The corresponding Edgeworth box is represented in Figure 19.C.2. We see that at any point of the Pareto set the common marginal rate of substitution is smaller than the ratio of probabilities (see Exercise 19.C.2). Hence at an equilibrium we must have $p_1/p_2 < \pi_1/\pi_2$, or $p_1/\pi_1 < p_2/\pi_2$. If, say, $\pi_1 = \pi_2 = \frac{1}{2}$, then $p_1 < p_2$: The price of one contingent unit of consumption is larger for the state for which the consumption good is scarcer. This constitutes the simplest version of a powerful theme of finance theory: that contingent instruments (in our case, a unit of contingent consumption) are comparatively more valuable if their returns (in our case, the amount of consumption they give in the different states) are negatively correlated with the "market return" (in our case, the random variable representing the aggregate initial endowment). ■

**Figure 19.C.2**

There is aggregate risk: p_1/p_2 negatively correlated with total endowment of commodity 1.

19.D Sequential Trade

The Arrow–Debreu framework provides a remarkable illustration of the power of general equilibrium theory. Yet, it is hardly realistic. Indeed, at an Arrow–Debreu equilibrium all trade takes place simultaneously and before the uncertainty is resolved. Trade, so to speak, is a one-shot affair. In reality, however, trade takes place to a large extent sequentially over time, and frequently as a consequence of information disclosures. The aim of this section is to introduce a first model of sequential trade and show that Arrow–Debreu equilibria can be reinterpreted by means of trading processes that actually unfold through time.

To be as simple as possible we consider only exchange economies (see Section 19.G for some discussion of production). In addition, we take $X_i = \mathbb{R}_+^{LS}$ for every i . To begin with, we assume that there are two dates, $t = 0$ and $t = 1$, that there is no information whatsoever at $t = 0$, and that the uncertainty has resolved completely at $t = 1$. Thus, the date-event tree is as in Figure 19.B.1. Again for simplicity, we assume that there is no consumption at $t = 0$. (We refer to Exercise 19.D.3 for the more general situation.)

Suppose that markets for the LS possible contingent commodities are set up at $t = 0$, and that $(x_1^*, \dots, x_I^*) \in \mathbb{R}^{LSI}$ is an Arrow–Debreu equilibrium allocation with prices $(p_{11}, \dots, p_{LS}) \in \mathbb{R}^{LS}$. Recall that these markets are for delivery of goods at $t = 1$ (they are commonly called *forward markets*). When period $t = 1$ arrives, a state of the world s is revealed, contracts are executed, and every consumer i receives $x_{si}^* = (x_{1si}^*, \dots, x_{LSi}^*) \in \mathbb{R}^L$. Imagine now that, after this but before the actual consumption of x_{si}^* , markets for the L physical goods were to open at $t = 1$ (these are called *spot markets*). Would there be any incentive to trade in these markets? The answer is “no.” To see why, suppose that there were potential gains from trade among the consumers. That is, that there were $x_{si} = (x_{1si}, \dots, x_{LSi})$ for $i = 1, \dots, I$, such that $\sum_i x_{si} \leq \sum_i \omega_{si}$ and $(x_{1i}^*, \dots, x_{si}, \dots, x_{Si}^*) \succsim_i (x_{1i}^*, \dots, x_{si}^*, \dots, x_{Si}^*)$ for all i , with at least one preference strict. It then follows from the definition of Pareto optimality that the Arrow–Debreu equilibrium allocation $(x_1^*, \dots, x_I^*) \in \mathbb{R}^{LSI}$ is not Pareto optimal, contradicting the conclusion of the first welfare theorem.⁵ In summary, at $t = 0$ the consumers can trade directly to an overall Pareto optimal allocation; hence there is no reason for further trade to take place. In other words, ex ante Pareto optimality implies ex post Pareto optimality and thus no ex post trade.

Matters are different if not all the LS contingent commodity markets are available at $t = 0$. Then the initial trade to a Pareto optimal allocation may not be feasible and it is quite possible that ex post (i.e., after the revelation of the state s) the resulting consumption allocation is not Pareto optimal. There would then be an incentive to reopen the markets and retrade.

5. Alternatively, consider the Arrow–Debreu equilibrium prices for the L contingent commodities corresponding to state s : $p_s = (p_{1s}, \dots, p_{Ls})$. Then p_s , viewed as a system of spot prices at s , induces, for the initial endowment vector $(x_{s1}^*, \dots, x_{sI}^*)$, a null excess demand for all traders (and therefore clears markets). Indeed, if $U_i(x_{1i}, \dots, x_{Si})$ is a utility function for $\succsim_i$ and $(x_{1i}^*, \dots, x_{Si}^*) \in \mathbb{R}^{LS}$ maximizes $U_i(x_{1i}, \dots, x_{Si})$ subject to $\sum_s p_s \cdot (x_{si} - \omega_{si}) \leq 0$, then, for any particular s , x_{is}^* maximizes $U_i(x_{1i}^*, \dots, x_{si}, \dots, x_{Si}^*)$ subject to $p_s \cdot (x_{si} - \omega_{si}) \leq p_s \cdot (x_{si}^* - \omega_{si})$, that is, subject to $p_s \cdot (x_{si} - x_{si}^*) \leq 0$.

A most interesting possibility, first observed by Arrow (1953), is that, even if not all the contingent commodities are available at $t = 0$, it may still be the case under some conditions that the retrading possibilities at $t = 1$ guarantee that Pareto optimality is reached, nevertheless. That is, the possibility of ex post trade can make up for an absence of some ex ante markets. In what follows, we shall verify that this is the case whenever at least one physical commodity can be traded contingently at $t = 0$ if, in addition, spot markets occur at $t = 1$ and the spot equilibrium prices are correctly anticipated at $t = 0$. The intuition for this result is reasonably straightforward: if spot trade can occur within each state, then the only task remaining at $t = 0$ is to transfer the consumer's overall purchasing power efficiently across states. This can be accomplished using contingent trade in a single commodity. By such a procedure we are able to reduce the number of required forward markets for LS to S .

Let us be more specific. At $t = 0$ consumers have *expectations* regarding the spot prices prevailing at $t = 1$ for each possible state $s \in S$. Denote the price vector expected to prevail in state s spot market by $p_s \in \mathbb{R}^L$, and the overall expectation vector⁶ by $p = (p_1, \dots, p_S) \in \mathbb{R}^{LS}$. Suppose that, in addition, at date $t = 0$ there is trade in the S contingent commodities denoted by 11 to 1S; that is, there is contingent trade only in the physical good with the label 1. We denote the vector of prices for these contingent commodities traded at $t = 0$ by $q = (q_1, \dots, q_S) \in \mathbb{R}^S$.

Faced with prices $q \in \mathbb{R}^S$ at $t = 0$ and expected spot prices $(p_1, \dots, p_S) \in \mathbb{R}^{LS}$ at $t = 1$, every consumer i formulates a consumption, or trading, plan $(z_{1i}, \dots, z_{Si}) \in \mathbb{R}^S$ for contingent commodities at $t = 0$, as well as a set of spot market consumption plans $(x_{1i}, \dots, x_{Si}) \in \mathbb{R}^{LS}$ for the different states that may occur at $t = 1$. Of course, these plans must satisfy a budget constraint. Let $U_i(\cdot)$ be a utility function for $\succeq_i$. Then the problem of consumer i can be expressed formally as

$$\begin{aligned} & \text{Max} && U_i(x_{1i}, \dots, x_{Si}) && (19.D.1) \\ & (x_{1i}, \dots, x_{Si}) \in \mathbb{R}^{LS} \\ & (z_{1i}, \dots, z_{Si}) \in \mathbb{R}^S \\ & \text{s.t.} && \text{(i) } \sum_s q_s z_{si} \leq 0, \\ & && \text{(ii) } p_s \cdot x_{si} \leq p_s \cdot \omega_{si} + p_{1s} z_{si} \quad \text{for every } s. \end{aligned}$$

Restriction (i) is the budget constraint corresponding to trade at $t = 0$. The family of restrictions (ii) are the budget constraints for the different spot markets. Note that the value of wealth at a state s is composed of two parts: the market value of the initial endowments, $p_s \cdot \omega_{si}$, and the market value of the amounts z_{si} of good 1 bought or sold forward at $t = 0$. Observe that we are not imposing any restriction on the sign or the magnitude of z_{si} . If $z_{si} < -\omega_{1si}$ then one says that at $t = 0$ consumer i is selling good 1 *short*. This is because he is selling at $t = 0$, contingent on state s occurring, more than he has at $t = 1$ if s occurs. Hence, if s occurs he will actually have to buy in the spot market the extra amount of the first good required for the fulfillment of his commitments. The possibility of selling short is, however, indirectly

6. In principle, expectations could differ across consumers, but under the assumption of correct expectations (soon to be introduced) they will not.

limited by the fact that consumption, and therefore ex post wealth, must be nonnegative for every s .⁷

To define an appropriate notion of sequential trade we shall impose a key condition: Consumers' expectations must be *self-fulfilled*, or *rational*; that is, we require that consumers' expectations of the prices that will clear the spot markets for the different states s do actually clear them once date $t = 1$ has arrived and a state s is revealed.

Definition 19.D.1: A collection formed by a price vector $q = (q_1, \dots, q_S) \in \mathbb{R}^S$ for contingent first good commodities at $t = 0$, a spot price vector

$$p_s = (p_{1s}, \dots, p_{Ls}) \in \mathbb{R}^L$$

for every s , and, for every consumer i , consumption plans $z_i^* = (z_{1i}^*, \dots, z_{Si}^*) \in \mathbb{R}^S$ at $t = 0$ and $x_i^* = (x_{1i}^*, \dots, x_{Li}^*) \in \mathbb{R}^{LS}$ at $t = 1$ constitutes a *Radner equilibrium* [see Radner (1982)] if:

- (i) For every i , the consumption plans z_i^*, x_i^* solve problem (19.D.1).
- (ii) $\sum_i z_{si}^* \leq 0$ and $\sum_i x_{si}^* \leq \sum_i \omega_{si}$ for every s .

At a Radner equilibrium, trade takes place through time and, in contrast to the Arrow–Debreu setting, economic agents face a *sequence of budget sets*, one at each date-state (more generally, at every date-event).

We can see from an examination of problem (19.D.1) that all the budget constraints are homogeneous of degree zero with respect to prices. This means that the budget sets remain unaltered if the price of one physical commodity in each date-state (that is, one price for every budget set) is arbitrarily normalized to equal 1. It is natural to choose the first commodity and to put $p_{1s} = 1$ for every s , so that a unit of the s contingent commodity then pays off 1 dollar in state s .⁸ Note that this still leaves one degree of freedom, that corresponding to the forward trades at date 0 (so we could put $q_1 = 1$, or perhaps $\sum_s q_s = 1$).

In Proposition 19.D.1, which is the key result of this section, we show that for this model the set of Arrow–Debreu equilibrium allocations (induced by the arrangement of one-shot trade in LS contingent commodities) and the set of Radner equilibrium allocations (induced by contingent trade in only one commodity, sequentially followed by spot trade) are identical.

7. Observe also that we have taken the wealth at $t = 0$ to be zero (that is, there are no initial endowments of the contingent commodities). This is simply a convention. Suppose, for example, that we regard ω_{1si} , the amount of good 1 available at $t = 1$ in state s , as the amount of the s contingent commodity that i owns at $t = 0$ (to avoid double counting, the initial endowment of commodity 1 in the spot market s at $t = 1$ should simultaneously be put to zero). The budget constraints are then: (i) $\sum_s q_s(z'_{si} - \omega_{1si}) \leq 0$ and (ii) $p_s \cdot x_{si} \leq \sum_{\ell \neq 1} p_{\ell s} \omega_{\ell 1} + p_{1s} z'_{si}$ for every s . But letting $z'_{si} = z_{si} + \omega_{1si}$, we see that these are exactly the constraints of (19.D.1).

8. It follows from the possibility of making this normalization that, without loss of generality, we could as well suppose that our contingent commodity pays directly in dollars (see Exercise 19.D.1 for more on this).

Proposition 19.D.1: We have:

(i) If the allocation $x^* \in \mathbb{R}^{LSI}$ and the contingent commodities price vector $(p_1, \dots, p_S) \in \mathbb{R}_{++}^{LS}$ constitute an Arrow–Debreu equilibrium, then there are prices $q \in \mathbb{R}_{++}^S$ for contingent first good commodities and consumption plans for these commodities $z^* = (z_1^*, \dots, z_I^*) \in \mathbb{R}^{SI}$ such that the consumptions plans x^* , z^* , the prices q , and the spot prices $(p_1, \dots, p_S)$ constitute a Radner equilibrium.

(ii) Conversely, if the consumption plans $x^* \in \mathbb{R}^{LSI}$, $z^* \in \mathbb{R}^{SI}$ and prices $q \in \mathbb{R}_{++}^S$, $(p_1, \dots, p_S) \in \mathbb{R}_{++}^{LS}$ constitute a Radner equilibrium, then there are multipliers $(\mu_1, \dots, \mu_S) \in \mathbb{R}_{++}^S$ such that the allocation x^* and the contingent commodities price vector $(\mu_1 p_1, \dots, \mu_S p_S) \in \mathbb{R}_{++}^{LS}$ constitute an Arrow–Debreu equilibrium. (The multiplier μ_s is interpreted as the value, at $t = 0$, of a dollar at $t = 1$ and state s .)

Proof: (i) It is natural to let $q_s = p_{1s}$ for every s . With this we claim that, for every consumer i , the budget set of the Arrow–Debreu problem,

$$B_i^{AD} = \{(x_{1i}, \dots, x_{Si}) \in \mathbb{R}_+^{LS} : \sum_s p_s \cdot (x_{si} - \omega_{si}) \leq 0\},$$

is identical to the budget set of the Radner problem,

$$B_i^R = \{(x_{1i}, \dots, x_{Si}) \in \mathbb{R}_+^{LS} : \text{there are } (z_{1i}, \dots, z_{Si}) \text{ such that } \sum_s q_s z_{si} \leq 0 \text{ and } p_s \cdot (x_{si} - \omega_{si}) \leq p_{1s} z_{si} \text{ for every } s\}.$$

To see this, suppose that $x_i = (x_{1i}, \dots, x_{Si}) \in B_i^{AD}$. For every s , denote $z_{si} = (1/p_{1s})p_s \cdot (x_{si} - \omega_{si})$. Then $\sum_s q_s z_{si} = \sum_s p_{1s} z_{si} = \sum_s p_s \cdot (x_{si} - \omega_{si}) \leq 0$ and $p_s \cdot (x_{si} - \omega_{si}) = p_{1s} z_{si}$ for every s . Hence, $x_i \in B_i^R$. Conversely, suppose that $x_i = (x_{1i}, \dots, x_{Si}) \in B_i^R$, that is, for some $(z_{1i}, \dots, z_{Si})$ we have $\sum_s q_s z_{si} \leq 0$ and $p_s \cdot (x_{si} - \omega_{si}) \leq p_{1s} z_{si}$ for every s . Summing over s gives $\sum_s p_s \cdot (x_{si} - \omega_{si}) \leq \sum_s p_{1s} z_{si} = \sum_s q_s z_{si} \leq 0$. Hence, $x_i \in B_i^{AD}$.

We conclude that our Arrow–Debreu equilibrium allocation is also a Radner equilibrium allocation supported by $q = (p_{11}, \dots, p_{1S}) \in \mathbb{R}^S$, the spot prices $(p_1, \dots, p_S)$, and the contingent trades $(z_{1i}^*, \dots, z_{Si}^*) \in \mathbb{R}^S$ defined by $z_{si}^* = (1/p_{1s})p_s \cdot (x_{si}^* - \omega_{si})$. Note that the contingent markets clear since, for every s , $\sum_i z_{si}^* = (1/p_{1s})p_s \cdot [\sum_i (x_{si}^* - \omega_{si})] \leq 0$.

(ii) Choose μ_s so that $\mu_s p_{1s} = q_s$. Then we can rewrite the Radner budget set of every consumer i as

$$B_i^R = \{(x_{1i}, \dots, x_{Si}) \in \mathbb{R}^{LS} : \text{there are } (z_{1i}, \dots, z_{Si}) \text{ such that } \sum_s q_s z_{si} \leq 0 \text{ and } \mu_s p_s \cdot (x_{si} - \omega_{si}) \leq q_s z_{si} \text{ for every } s\}.$$

But from this we can proceed as we did in part (i) and rewrite the constraints, and therefore the budget set, in the Arrow–Debreu form:

$$B_i^R = B_i^{AD} = \{(x_{1i}, \dots, x_{Si}) \in \mathbb{R}^{LS} : \sum_s \mu_s p_s \cdot (x_{si} - \omega_{si}) \leq 0\}.$$

Hence, the consumption plan x_i^* is also preference maximizing in the budget set B_i^{AD} . Since this is true for every consumer i , we conclude that the price vector $(\mu_1 p_1, \dots, \mu_S p_S) \in \mathbb{R}^{LS}$ clears the markets for the LS contingent commodities. ■

Example 19.D.1: Consider a two-good, two-state, two-consumer pure exchange economy. Suppose that the two states are equally likely and that every consumer has the same, state-independent, Bernoulli utility function $u(x_{si})$. The consumers differ only in their initial endowments. The aggregate endowment vectors in the two states

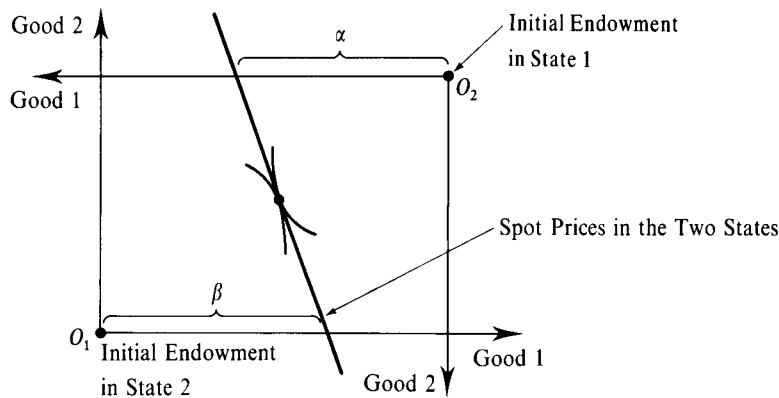


Figure 19.D.1
Reaching the Arrow–Debreu equilibrium by means of contingent trade in the first good only.

are the same; however, endowments are distributed so that consumer 1 gets everything in state 1 and consumer 2 gets everything in state 2. (See Figure 19.D.1.)

By the symmetry of the problem, at an Arrow–Debreu equilibrium each consumer gets, in each state, half of the total endowment of each good. In Figure 19.D.1, we indicate how these consumptions will be reached by means of contingent trade in the first commodity and spot markets. The spot prices will be the same in the two states. The first consumer will sell an amount α of the first good contingent on the occurrence of the first state and will in exchange buy an amount β of the same good contingent on the second state. (You are asked to provide the details in Exercise 19.D.2.) ■

It is important to emphasize that, although the concept of Radner equilibrium cuts down the number of contingent commodities required to attain optimality (from LS to S), this reduction is not obtained free of charge. With the smaller number of forward contracts, the correct anticipation of future spot prices becomes crucial.

Up to this point we have discussed the sequential implementation of an Arrow–Debreu equilibrium when there are two dates,⁹ that is, for the date–event tree of Figure 19.B.1. Except for notational complications, the same ideas carry over to a tree such as that in Figure 19.B.2 where there are $T + 1$ periods and information is released gradually. (See the small-type discussion at the end of Section 19.B for basic concepts and notation.) We would then have spot markets at every admissible date–event pair tE (i.e., those tE where $E \in \mathcal{S}_t$, the information partition at t). With H the set of basic physical commodities, we denote the spot prices by $p_{tE} \in \mathbb{R}^H$. At every tE we could also have trade for the contingent delivery of physical good 1 at each of the successor date–events to tE . Denote by $q_{tE}(t + 1, E')$ the price at tE of one unit of good 1 delivered at $t + 1$ if event E' is revealed (of course, we require $E' \in \mathcal{S}_{t+1}$ and $E' \subset E$). The problem of the consumer consists of forming utility-maximizing plans by choosing, at every admissible tE , a vector of consumption of goods $x_{tEi} \in \mathbb{R}_+^H$ and, for every successor $(t + 1, E')$, a contingent trade $z_{tEi}(t + 1, E')$ of good 1 deliverable at $(t + 1, E')$. Overall, the budget constraint to be satisfied at tE is

$$p_{tE} \cdot x_{tEi} + \sum_{\{E' \in \mathcal{S}_{t+1}: E' \subset E\}} q_{tE}(t + 1, E') z_{tEi}(t + 1, E') \leq p_{tE} \cdot \omega_{tEi} + p_{1tE} z_{t-1, E^-}(t, E)$$

where E^- is the event at the date $t - 1$ predecessor to event E at t .

9. To be as simple as possible, we have also assumed that there is no consumption at $t = 0$.

One can then proceed to define a corresponding concept of Radner equilibrium and to show that the Arrow–Debreu equilibrium allocations for the model with $H(T + 1)S$ contingent commodity markets¹⁰ at $t = 0$ are the same as the Radner equilibrium allocations obtained from a model with sequential trade in which, at each date-event, consumers trade only current goods and contingent claims for delivery of good 1 at successor nodes. Exercises 19.D.3 and 19.D.4 discuss this topic further.

19.E Asset Markets

The S contingent commodities studied in the previous section serve the purpose of transferring wealth across the states of the world that will be revealed in the future. They are, however, only theoretical constructs that rarely have exact counterparts in reality. Nevertheless, in reality there are *assets*, or *securities*, that to some extent perform the wealth-transferring role that we have assigned to the contingent commodities. It is therefore important to develop a theoretical structure that allows us to study the functioning of these asset markets. We accomplish the task in this section by extending the formal notion of a contingent commodity and then generalizing the theory of Radner equilibrium to the extended environment.¹¹

We begin again with the simplest situation, in which we have two dates, $t = 0$ and $t = 1$, and all the information is revealed at $t = 1$. Further, for notational simplicity we assume that consumption takes place only at $t = 1$.

We view an asset, or, more precisely, a unit of an asset, as a title to receive either physical goods or dollars at $t = 1$ in amounts that may depend on which state occurs.¹² The payoffs of an asset are known as its *returns*. If the returns are in physical goods, the asset is called *real* (a durable piece of machinery or a futures contract for the delivery of copper would be examples). If they are in paper money, they are called *financial* (a government bond, for example). Mixed cases are also possible. Here we deal only with the real case and, moreover, to save on notation we assume that the returns of assets are only in amounts of physical good 1.¹³ It is then convenient to normalize the spot price of that good to be 1 in every state, so that, in effect, we are using it as numeraire.

Definition 19.E.1: A unit of an *asset*, or *security*, is a title to receive an amount r_s of good 1 at date $t = 1$ if state s occurs. An asset is therefore characterized by its *return vector* $r = (r_1, \dots, r_S) \in \mathbb{R}^S$.

10. A contingent commodity is a promise to deliver a unit of physical commodity h at date t if state s occurs. Recall from Section 19.B that the consumption sets have to be defined imbedding in them the information measurability restrictions, that is, making sure that at date t no consumption is dependent on information not yet available.

11. See Radner (1982) and Kreps (1979) [complemented by Marimon (1987)] for treatments in the spirit of this section.

12. As usual “title to receive” means “duty to deliver” if the amount is negative. Although negative returns present no particular difficulty, we will avoid them.

13. This assumption also has an important simplifying feature: At any given state the returns of all assets are in units of the same physical good. Therefore, the relative spot prices of the various physical goods in any given state do not affect the relative returns of the different assets in that state.

Example 19.E.1: Examples of assets include the following:

- (i) $r = (1, \dots, 1)$. This asset promises the future noncontingent delivery of one unit of good 1. Its real-world counterparts are the markets for *commodity futures*. In the special case where there is a single consumption good (i.e., $L = 1$), we call this asset the *safe* (or *riskless*) *asset*. It is important to realize that with more than one physical good a futures contract is not riskless: its return in terms of purchasing power depends on the spot prices of all the goods.¹⁴
- (ii) $r = (0, \dots, 0, 1, 0, \dots, 0)$. This asset pays one unit of good 1 if and only if a certain state occurs. These were the assets considered in Section 19.D. In the current theoretical setting they are often called *Arrow securities*.
- (iii) $r = (1, 2, 1, 2, \dots, 1, 2)$. This asset pays one unit unconditionally and, in addition, another unit in even-labeled states. ■

Example 19.E.2: Options. This is an example of a so-called *derivative asset*, that is, of an asset whose returns are somehow derived from the returns of another asset. Suppose there is a *primary asset* with return vector $r \in \mathbb{R}^S$. Then a (*European*) *call option* on the primary asset at the *strike price* $c \in \mathbb{R}$ is itself an asset. A unit of this asset gives the option to buy, *after the state is revealed* (but before the returns are paid), a unit of the primary asset at price c (the price c is in units of the “numeraire,” that is, of good 1).

What is the return vector $r(c)$ of the option? In a given state s , the option will be exercised if and only if $r_s > c$ (we neglect the case $r_s = c$). Hence

$$r(c) = (\text{Max}\{0, r_1 - c\}, \dots, \text{Max}\{0, r_S - c\}).$$

For a primary asset with returns $r = (4, 3, 2, 1)$ specific examples are

$$\begin{aligned} r(3.5) &= (.5, 0, 0, 0), \\ r(2.5) &= (1.5, .5, 0, 0), \\ r(1.5) &= (2.5, 1.5, .5, 0). \quad \blacksquare \end{aligned}$$

We proceed to extend the analysis of Section 19.D by assuming that there is a given set of assets, known as an *asset structure*, and that these assets can be freely traded at date $t = 0$. We postpone to the next section a discussion of the important issue of the origin of the particular set of assets. Each asset k is characterized by a vector of returns $r_k \in \mathbb{R}^S$. The number of assets is K . As before, we assume that there are no initial endowments of assets and that short sales are possible. The price vector for the assets traded at $t = 0$ is denoted $q = (q_1, \dots, q_K)$. A vector of trades in these assets, denoted by $z = (z_1, \dots, z_K) \in \mathbb{R}^K$, is called a *portfolio*.

The next step is to generalize the definition of a Radner equilibrium to the current environment. In Definition 19.E.2, $U_i(\cdot)$ is a utility function for the preferences $\succeq_i$ of consumer i over consumption plans $(x_{1i}, \dots, x_{Si}) \in \mathbb{R}_+^{LS}$.

14. Strictly speaking, for the term “riskless” to be meaningful we need, in addition to $L = 1$, that utility functions be uniform across states.

Definition 19.E.2: A collection formed by a price vector $q = (q_1, \dots, q_K) \in \mathbb{R}^K$ for assets traded at $t = 0$, a spot price vector $p_s = (p_{1s}, \dots, p_{Ls}) \in \mathbb{R}^L$ for every s , and, for every consumer i , portfolio plans $z_i^* = (z_{1i}^*, \dots, z_{Ki}^*) \in \mathbb{R}^K$ at $t = 0$ and consumption plans $x_i^* = (x_{1i}^*, \dots, x_{Si}^*) \in \mathbb{R}^{LS}$ at $t = 1$ constitutes a *Radner equilibrium* if:

(i) For every i , the consumption plans z_i^*, x_i^* solve the problem

$$\begin{aligned} \text{Max} \quad & U_i(x_{1i}, \dots, x_{Si}) \\ & (x_{1i}, \dots, x_{Si}) \in \mathbb{R}_+^{LS} \\ & (z_{1i}, \dots, z_{Ki}) \in \mathbb{R}^K \\ \text{s.t.} \quad & \text{(a) } \sum_k q_k \cdot z_{ki} \leq 0 \\ & \text{(b) } p_s \cdot x_{si} \leq p_s \cdot \omega_{si} + \sum_k p_{1s} z_{ki} r_{sk} \quad \text{for every } s. \end{aligned}$$

(ii) $\sum_i z_{ki}^* \leq 0$ and $\sum_i x_{si}^* \leq \sum_i \omega_{si}$ for every k and s .

In the budget set of Definition 19.E.2, the wealth of consumer i at state s is the sum of the spot value of his initial endowment and the spot value of the return of his portfolio. Note that, without loss of generality, we can put $p_{1s} = 1$ for all s . From now on we will do so. It is convenient at this point to introduce the concept the *return matrix* R . This is an $S \times K$ matrix whose k th column is the return vector of the k th asset. Hence, its generic sk entry is r_{sk} , the return of asset k in state s . With this notation, the budget constraint of consumer i becomes

$$B_i(p, q, B) = \left\{ x \in \mathbb{R}_+^{LS} : \text{for some portfolio } z_i \in \mathbb{R}^K \text{ we have } q \cdot z_i \leq 0 \text{ and} \right. \\ \left. \begin{pmatrix} p_1 \cdot (x_{1i} - \omega_{1i}) \\ \vdots \\ p_s \cdot (x_{Si} - \omega_{Si}) \end{pmatrix} \leq \begin{bmatrix} r_{11}, \dots, r_{1K} \\ \vdots \\ r_{S1}, \dots, r_{SK} \end{bmatrix} z_i = R z_i \right\}$$

We now present a very important implication, rich in ramifications, of the assumption that unlimited short sales are possible. Namely, we will establish that knowledge of the return matrix R suffices to place significant restrictions on the asset price vector $q = (q_1, \dots, q_K)$ that could conceivably arise at equilibrium.

Proposition 19.E.1: Assume that every return vector is nonnegative and nonzero; that is, $r_k \geq 0$ and $r_k \neq 0$ for all k .¹⁵ Then, for every (column) vector $q \in \mathbb{R}^K$ of asset prices arising in a Radner equilibrium, we can find multipliers $\mu = (\mu_1, \dots, \mu_S) \geq 0$, such that $q_k = \sum_s \mu_s r_{sk}$ for all k (in matrix notation, $q^T = \mu \cdot R$).

In words, Proposition 19.E.1 says that we can assign values $(\mu_1, \dots, \mu_S)$ to units of wealth in the different states so that the price of a unit of asset k is simply equal to the sum, in value terms, of the returns across states.¹⁶ Because this is an important

15. This assumption can be weakened substantially.

16. As we shall see shortly, the value μ_s can also be interpreted as the implicit price of the state-contingent commodity that pays one unit of good 1 if state s occurs and nothing otherwise.

result, we shall give two proofs of it. The first, which we give in small type, is based on convexity theory and uses only one implication of equilibrium: the fact that q must be *arbitrage free* (we will give a definition of this concept shortly). The second proof uses the first-order conditions of the utility maximization problem and provides further insight into the nature of the multipliers.

Proof 1 of Proposition 19.E.1: Call the system $q \in \mathbb{R}^K$ of asset prices *arbitrage free* if there is no portfolio $z = (z_1, \dots, z_K)$ such that $q \cdot z \leq 0$, $Rz \geq 0$, and $Rz \neq 0$. In words, there is no portfolio that is budgetarily feasible and that yields a nonnegative return in *every* state and a strictly positive return in some state. Note that whether an asset price vector is arbitrage free or not depends only on the returns of the assets and not on preferences.

If, as usual, we assume that preferences are strongly monotone, then an equilibrium asset price vector $q \in \mathbb{R}^K$ must be arbitrage free: if it were not, it would be possible to increase utility merely by adding to any current portfolio a portfolio yielding an arbitrage opportunity. Because there are no restrictions on short sales this addition is always feasible.

In Lemma 19.E.1 we establish a result which in view of the observation just made is formally stronger than the statement of Proposition 19.E.1.

Lemma 19.E.1: If the asset price vector $q \in \mathbb{R}^K$ is arbitrage free, then there is a vector of multipliers $\mu = (\mu_1, \dots, \mu_S) \geq 0$ satisfying $q^T = \mu \cdot R$.

Proof of Lemma 19.E.1: Note to begin with that since we deal only with assets having nonnegative, nonzero returns, an arbitrage-free price vector q must have $q_k > 0$ for every k . Also, without loss of generality, we assume that no row of the return matrix R has all of its entries equal to zero.¹⁷

Given an arbitrage-free asset price vector $q \in \mathbb{R}^K$, consider the convex set

$$V = \{v \in \mathbb{R}^S: v = Rz \text{ for some } z \in \mathbb{R}^K \text{ with } q \cdot z = 0\}.$$

The arbitrage freeness of q implies that $V \cap \{\mathbb{R}_+^S \setminus \{0\}\} = \emptyset$. Since both V and $\mathbb{R}_+^S \setminus \{0\}$ are convex sets and the origin belongs to V , we can apply the separating hyperplane theorem (see Section M.G of the Mathematical Appendix) to obtain a nonzero vector $\mu' = (\mu'_1, \dots, \mu'_S)$ such that $\mu' \cdot v \leq 0$ for any $v \in V$ and $\mu' \cdot v \geq 0$ for any $v \in \mathbb{R}_+^S$. Note that it must be that $\mu' \geq 0$. Moreover, because $v \in V$ implies $-v \in V$, it follows that $\mu' \cdot v = 0$ for any $v \in V$. Figure 19.E.1(a) depicts this construction for the two-state case.

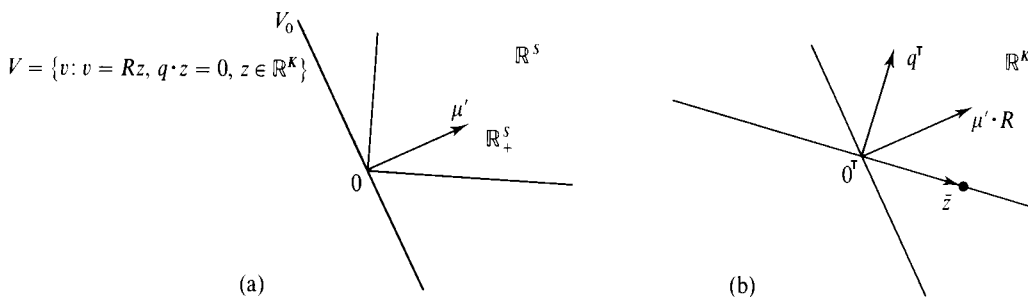


Figure 19.E.1

(a) Construction of the no-arbitrage weights.
(b) Existence of an inadmissible $\bar{z}$ if q^T is not proportional to $\mu' \cdot R$.

17. If there is such a row, set μ_s arbitrarily for the state s corresponding to that row, drop s from the list of states, and proceed with the remaining states.

We now argue that the row vector q^T must be proportional to the row vector $\mu' \cdot R \in \mathbb{R}^K$. The entries of μ' and of R are all nonnegative and no row of R is null. Therefore $\mu' \cdot R \geq 0^T$ and $\mu' \cdot R \neq 0^T$. If q^T is not proportional to $\mu' \cdot R$ then we can find $\bar{z} \in \mathbb{R}^K$ such that $q \cdot \bar{z} = 0$ and $\mu' \cdot R\bar{z} > 0$ [see Figure 19.E.1(b)]. But letting $v = R\bar{z}$, we would then have $v \in V$ and $\mu' \cdot v \neq 0$, which we have just seen cannot happen. Hence q^T must be proportional to $\mu' \cdot R$; that is, $q^T = \alpha \mu' \cdot R$ for some real number $\alpha > 0$. Letting $\mu = \alpha \mu'$, we have the conclusion of the lemma. ■

As we have already argued, if short sales of assets are possible and preferences are strongly monotone (e.g., if preferences admit an expected utility representation with strictly positive subjective probabilities for the states), then equilibrium asset prices must be arbitrage free and, therefore, Proposition 19.E.1 follows from Lemma 19.E.1. ■

Proof 2 of Proposition 19.E.1: For this proof we assume that preferences are represented by utility functions of the expected utility form $U_i(x_{1i}, \dots, x_{Si}) = \sum_s \pi_{si} u_{si}(x_{si})$ and that the Bernoulli utilities $u_{si}(\cdot)$ are concave, strictly increasing, and differentiable. We denote by $v_{si}(p_s, w_{si})$ the indirect utility function [derived from $u_{si}(\cdot)$] of consumer i in state s .

Suppose that in the Radner equilibrium with asset prices $q = (q_1, \dots, q_K)$ the equilibrium spot prices are $p = (p_1, \dots, p_S) \in \mathbb{R}^{LS}$. Because unlimited short sales are possible, the optimal portfolio choice $z_i^* \in \mathbb{R}^K$ of any consumer i is necessarily interior and, denoting $w_{si}^* = p_s \cdot \omega_{si} + \sum_k r_{sk} z_{ki}^*$, it must satisfy the following first-order conditions for some $\alpha_i > 0$:

$$\alpha_i q_k = \sum_s \pi_{si} \frac{\partial v_{si}(p_s, w_{si}^*)}{\partial w_{si}} r_{sk} \quad \text{for every } k = 1, \dots, K.$$

That is, the vector of expected marginal utilities of the K assets must be proportional to the vector of asset prices.¹⁸ With this we have attained our result, since by taking

$$\mu_{si} = \frac{\pi_{si}}{\alpha_i} \frac{\partial v_{si}(p_s, w_{si}^*)}{\partial w_{si}}$$

we have $q_j = \sum_s \mu_{si} r_{sj}$. Hence, we could determine $\mu = (\mu_1, \dots, \mu_S)$ by choosing any consumer i and letting $\mu_s = \mu_{si}$, the marginal utility of wealth at state s of consumer i weighted by π_{si}/α_i . The multiplier α_i is the Lagrange multiplier of the budget constraint at $t = 0$ and can therefore be viewed as the marginal utility of wealth at $t = 0$. Hence, for any consumer i , μ_{si} equals the ratio of the (expected) utility at $t = 0$ of one extra unit of wealth at $t = 1$ and state s , and the utility of one extra unit of wealth at $t = 0$. See Exercise 19.E.1 for more on this point. Note also that different consumers may lead to different $\mu_i = (\mu_{i1}, \dots, \mu_{iS})$ and therefore to different μ 's. The uniqueness of μ is assured only when $\text{rank } R = S$. ■

18. Recall that we always take $p_{1s} = 1$. Therefore, r_{sk} is the extra amount of wealth in state s derived from an extra unit of asset k . The proportionality factor α_i is the Lagrange multiplier of the problem

$$\begin{aligned} \text{Max} \quad & \sum_s \pi_{si} v_{si}(p_s, p_s \cdot \omega_{si} + \sum_k r_{sk} z_{ki}) \\ \text{s.t.} \quad & \sum_k q_k z_{ki} \leq 0. \end{aligned}$$

Example 19.E.3: Suppose that there is available an asset with noncontingent returns; for example, $r_1 = (1, \dots, 1)$. Normalize the price of this asset to be 1, that is, $q_1 = 1$. Then if $\mu = (\mu_1, \dots, \mu_S)$ is the vector of multipliers given by Proposition 19.E.1, we must have $\mu \geq 0$ and $\sum_s \mu_s = \mu \cdot r_1 = q_1 = 1$. For any other asset k we then obtain the intuitive conclusion that $q_k = \sum_s \mu_s r_{sk} \geq \text{Min}_s r_{sk}$ and, similarly, $q_k \leq \text{Max}_s r_{sk}$. ■

In Section 19.D, we proved that for the set of assets consisting of the S contingent markets in a single physical commodity we have an equivalence result between Arrow–Debreu and Radner equilibrium allocations (Proposition 19.D.1). We now generalize this result. In particular, we show that this equivalence holds for *any* family of S or more assets, *provided* that at least S of them have returns that are linearly independent (i.e., provided the effective number of assets is at least S). We begin with Definition 19.E.2.

Definition 19.E.3: An asset structure with an $S \times K$ return matrix R is *complete* if $\text{rank } R = S$, that is, if there is some subset of S assets with linearly independent returns.

Example 19.E.4: In the case of S contingent commodities discussed in Section 19.D, and also in Example 19.E.1(ii), the return matrix R is the $S \times S$ identity matrix. This is the canonical example of complete markets. But there are many other ways for a matrix to be nonsingular. Thus, with three states and three assets, we could have the return matrix

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix},$$

which has rank equal to 3, the number of states. ■

Example 19.E.5: Spanning through Options. Suppose that $S = 4$ and there is a primary asset with returns $r = (4, 3, 2, 1)$. We have seen in Example 19.E.2 that, for every strike price c , the option defined by c constitutes an asset with return vector $r(c) = (\text{Max } \{0, r_1 - c\}, \dots, \text{Max } \{0, r_4 - c\})$. Using options we can create a complete asset structure supported entirely on the primary asset r . For example, the return vectors $r(3.5)$, $r(2.5)$, $r(1.5)$, and r are linearly independent (the matrix R has all its entries below the diagonal equal to zero). Thus, the asset structure consisting of the primary asset plus three options with strike prices 3.5, 2.5, and 1.5 is complete. More generally, whenever the primary asset is such that $r_s \neq r_{s'}$ for all $s \neq s'$, it is possible to generate a complete asset structure by means of options (see Exercise 19.E.2). If $r_s = r_{s'}$ for some distinct s and s' , then we cannot do so: If the primary asset does not distinguish between two states, no derived asset can do so either. ■

To repeat, the importance of the concept of completeness derives from the fact that with it we can generalize Proposition 19.D.1. With a complete asset structure, economic agents are, in effect, unrestricted in their wealth transfers across states (except, of course, by their budget constraints). Therefore, at the equilibrium, their portfolio choices induce the same second-period consumptions as in the Arrow–Debreu equilibrium, and so full Pareto optimality is reached. This is the content of Proposition 19.E.2.

Proposition 19.E.2: Suppose that the asset structure is complete. Then:

- (i) If the consumption plans $x^* = (x_1^*, \dots, x_I^*) \in \mathbb{R}^{LSI}$ and the price vector $(p_1, \dots, p_S) \in \mathbb{R}_{++}^{LS}$

constitute an Arrow–Debreu equilibrium, then there are asset prices $q \in \mathbb{R}_{++}^K$ and portfolio plans $z^* = (z_1^*, \dots, z_I^*) \in \mathbb{R}^{KI}$ such that the consumption plans x^* , portfolio plans z^* , asset prices q , and spot prices $(p_1, \dots, p_S)$ constitute a Radner equilibrium.

- (ii) Conversely, if the consumption plans $x^* \in \mathbb{R}^{LSI}$, portfolio plans $z^* \in \mathbb{R}^{KI}$, and prices $q \in \mathbb{R}_{++}^K$, $(p_1, \dots, p_S) \in \mathbb{R}_{++}^{LS}$ constitute a Radner equilibrium, then there are multipliers $(\mu_1, \dots, \mu_S) \in \mathbb{R}_{++}^S$ such that the consumption plans x^* and the contingent commodities price vector $(\mu_1 p_1, \dots, \mu_S p_S) \in \mathbb{R}^{LS}$ constitute an Arrow–Debreu equilibrium. (The multiplier μ_s is interpreted as the value, at $t = 0$, of a dollar at $t = 1$ and state s ; recall that $p_{1s} = 1$.)

Proof: It is entirely similar to the proof of Proposition 19.D.1.

(i) Define $q_k = \sum_s p_{1s} r_{sk}$ for every k . Denote by Λ the $S \times S$ diagonal matrix whose s diagonal entry is p_{1s} . Then $q^T = e \cdot \Lambda R$, where $e \in \mathbb{R}^S$ is a column vector with all its entries equal to 1. For every i the (column) vector of wealth transfers across states (at the Arrow–Debreu equilibrium) is

$$m_i = (p_1 \cdot (x_{1i}^* - \omega_{1i}), \dots, p_S \cdot (x_{Si}^* - \omega_{Si}))^T.$$

We have $e \cdot m_i = 0$ for every i and $\sum_i m_i = 0$. By completeness, $\text{rank } \Lambda R = S$ and, therefore, we can find vectors $z_i^* \in \mathbb{R}^K$ such that $m_i = \Lambda R z_i^*$ for $i = 1, \dots, I - 1$. Letting

$$z_I^* = -(z_1^* + \dots + z_{I-1}^*)$$

we also have $m_I = -(m_1 + \dots + m_{I-1}) = \Lambda R z_I^*$. Therefore, for each i , the portfolio z_i^* allows consumer i to reach the Arrow–Debreu consumptions in the different states at the spot prices $(p_1, \dots, p_S)$. To verify budget feasibility note that $q \cdot z_i^* = e \cdot \Lambda R z_i^* = e \cdot m_i = 0$. In Exercise 19.E.3 you are asked to complete the proof by showing that the consumption and portfolio plans x_i^* and z_i^* are not just budget feasible but also utility maximizing in the budget set.

(ii) Assume, without loss of generality, that $p_{1s} = 1$ for all s . By Proposition 19.E.1 we have $q^T = \mu \cdot R$ for some arbitrage weights $\mu = (\mu_1, \dots, \mu_S)$. We show that x^* is an Arrow–Debreu equilibrium with respect to $(\mu_1 p_1, \dots, \mu_S p_S)$. To this effect, suppose that $x_i \in \mathbb{R}^{LS}$ satisfies the Arrow–Debreu single budget constraint, that is, $\sum_s \mu_s p_s \cdot (x_{si} - \omega_{si}) \leq 0$. Then by the completeness assumption there is $z_i \in \mathbb{R}^K$ such that $(p_1 \cdot (x_{1i} - \omega_{1i}), \dots, p_S \cdot (x_{Si} - \omega_{Si}))^T = R z_i$ and, therefore, $q \cdot z_i = \mu \cdot R z_i \leq 0$. Hence x_i also satisfies the budget constraints of the Radner equilibrium. Observe next that the Radner equilibrium consumption x_i^* is Arrow–Debreu budget feasible since $(p_1 \cdot (x_{1i}^* - \omega_{1i}), \dots, p_S \cdot (x_{Si}^* - \omega_{Si}))^T \leq R z_i^*$ and $q^T = \mu \cdot R$ yields

$$\sum_s \mu_s p_s \cdot (x_{si}^* - \omega_{si}) \leq \mu \cdot R z_i^* = q \cdot z_i^* \leq 0.$$

Therefore, x_i^* is utility maximizing in the Arrow–Debreu budget constraint. ■

It is important to realize that in discussing Radner equilibria what matters is not so much the particular asset structure but the linear space,

$$\text{Range } R = \{v \in \mathbb{R}^S: v = R z \text{ for some } z \in \mathbb{R}^K\} \subset \mathbb{R}^S,$$

the set of wealth vectors that can be *spanned* by means of the existing assets. It is quite possible for two different asset structures to give rise to the same linear space. Our next result, Proposition 19.E.3, tells us that, whenever this is so, the set of Radner equilibrium allocations for the two asset structures is the same.

Proposition 19.E.3: Suppose that the asset price vector $q \in \mathbb{R}^K$, the spot prices $p = (p_1, \dots, p_S) \in \mathbb{R}^{LS}$, the consumption plans $x^* = (x_1^*, \dots, x_I^*) \in \mathbb{R}_+^{LSI}$, and the portfolio plans $(z_1^*, \dots, z_I^*) \in \mathbb{R}^{KI}$ constitute a Radner equilibrium for an asset structure with $S \times K$ return matrix R . Let R' be the $S \times K'$ return matrix of a second asset structure. If $\text{Range } R' = \text{Range } R$, then x^* is still the consumption allocation of a Radner equilibrium in the economy with the second asset structure.

Proof: By Proposition 19.E.1, the asset prices satisfy the arbitrage condition $q^T = \mu \cdot R$, for some $\mu \in \mathbb{R}_+^S$. Denote $q' = [\mu \cdot R']^T$. We claim that if $\text{Range } R = \text{Range } R'$ then

$$B_i(p, q', R') = B_i(p, q, R) \quad \text{for every } i. \quad (19.E.1)$$

We show that if $x_i \in B_i(p, q, R)$ then $x_i \in B_i(p, q', R')$. To see this, let

$$(p_1 \cdot (x_{1i} - \omega_{1i}), \dots, p_S \cdot (x_{Si} - \omega_{Si}))^T \leq R z_i$$

and $q \cdot z_i \leq 0$. Since $\text{Range } R = \text{Range } R'$, we can find $z'_i \in \text{Range } R'$ such that $R z_i = R' z'_i$. But then $q' \cdot z'_i = \mu \cdot R' z'_i = \mu \cdot R z_i = q \cdot z_i \leq 0$, and therefore we can conclude that $x_i \in B_i(p, q', R')$. The converse statement [if $x_i \in B_i(p, q', R')$ then $x_i \in B_i(p, q, R)$] is proved in exactly the same way.

It follows from (19.E.1) that, for every consumer i , x_i^* is preference maximal in the budget set $B_i(p, q', R')$.

To argue that the asset prices q' , the spot prices $p = (p_1, \dots, p_S)$, and the consumption allocation x^* are part of a Radner equilibrium in the economy with an asset structure having return matrix R' , it suffices to find portfolios $(z'_1, \dots, z'_I) \in \mathbb{R}^{KI}$ such that, first, $\sum_i z'_i = 0$ and, second, for every consumer i , the vector of across-states wealth transfers

$$m_i = (p_1 \cdot (x_{1i}^* - \omega_{1i}), \dots, p_S \cdot (x_{Si}^* - \omega_{Si}))^T$$

satisfies $m_i = R' z'_i$. This is simple to accomplish. By strong monotonicity of preferences we have $m_i = R z_i^*$, for every i . Hence, $m_i \in \text{Range } R$ and therefore $m_i \in \text{Range } R'$ for every i . Choose then $z'_1, \dots, z'_{I-1}$ such that $m_i = R' z'_i$ for every $i = 1, \dots, I-1$. Finally, let $z'_I = -z'_1 - \dots - z'_{I-1}$. Then $\sum_i z'_i = 0$ and also

$$m_I = -(m_1 + \dots + m_{I-1}) = -R'(z'_1 + \dots + z'_{I-1}) = R' z'_I. \quad \blacksquare$$

One says that an asset is *redundant* if its deletion does not affect the linear space $\text{Range } R$ of spannable wealth transfers, that is, if its return vector is a linear combination of the return vectors of the remaining assets. It follows from Proposition 19.E.3 that the set of consumption allocations obtainable as part of a Radner equilibrium is not changed by the addition or deletion of a redundant asset. Another important fact is that a redundant asset can be priced merely by knowing the matrix of returns and the prices of the other assets.

Exercise 19.E.4: (Pricing by Arbitrage). Suppose that $r_3 = \alpha_1 r_1 + \alpha_2 r_2$. Show that at equilibrium we must have $q_3 = \alpha_1 q_1 + \alpha_2 q_2$. Recall that unlimited short sales are possible. (Assume also that the return vectors are nonnegative and nonzero.)

An implication of Exercise 19.E.4 is that, if the asset structure is complete, then we can deduce the prices of all assets from knowing the prices of a subset formed by S of them with linearly independent returns. A related way to see this is to note that from the prices of S assets with linearly independent returns we can uniquely deduce

the state multipliers $\mu = (\mu_1, \dots, \mu_S)$ of Proposition 19.E.1; indeed, for this we just have to solve a linear system of S independent equations in S unknowns. These multipliers can be interpreted as the (arbitrage) prices of the Arrow securities [Example 19.E.1(ii)]. Once we have these multipliers, we can obtain the price of any other asset k with return vector r_k as $q_k = \sum_s \mu_s r_{sk}$.

Example 19.E.6: Pricing an Option. Suppose that, with $S = 2$, there is an asset with uncontingent returns, say $r_1 = (1, 1)$ and a second asset $r_2 = (3 + \alpha, 1 - \alpha)$, with $\alpha > 0$. The asset prices are $q_1 = 1$ and q_2 . We now consider an option on the second asset that has strike price $c \in (1, 3)$. Then

$$r_2(c) = (3 + \alpha - c, 0) = \frac{3 + \alpha - c}{2 + 2\alpha} r_2 - \frac{(1 - \alpha)(3 + \alpha - c)}{2 + 2\alpha} r_1.$$

Therefore, the arbitrage price of the option (the only price compatible with equilibrium in the asset market) must be

$$q_2(c) = \frac{3 + \alpha - c}{2 + 2\alpha} [q_2 - (1 - \alpha)]. \quad (19.E.2)$$

An equivalent way to get the same formula is to note that, since the 2×2 return matrix R is nonsingular, the multipliers $\mu = (\mu_1, \mu_2)$ of Proposition 19.E.1 can be determined uniquely from $(1, q_2) = \mu \cdot R$. They are $\mu_1 = (q_2 - (1 - \alpha))/(2 + 2\alpha)$ and $\mu_2 = 1 - \mu_1$. But, again from Proposition 19.E.1, we have

$$q_2(c) = \mu \cdot r_2(c) = \mu_1(3 + \alpha - c),$$

which is precisely expression (19.E.2).

Note that if the prices of the two assets r_1 and r_2 are themselves arbitrage free, then we must have $3 + \alpha \geq q_2 \geq 1 - \alpha$ (recall Example 19.E.3). Therefore, we learn from formula (19.E.2) that $q_2(c)$ is nonnegative, decreasing in c , and increasing in q_2 .

We can also show that if the asset price q_2 stays constant but the dispersion parameter represented by α increases, then the option becomes more valuable. Suppose, in effect, that $\alpha' > \alpha$ and $r'_2(c)$, $r_2(c)$ are the corresponding returns of the option. Then $r_3 = r'_2(c) - r_2(c)$ is itself an asset with nonnegative returns. We can also price it by arbitrage from r_1 and r_2 to get a $q_3 \geq 0$ (typically, $q_3 > 0$). But then, again by arbitrage, $q'_2(c) = q_3 + q_2(c) \geq q_2(c)$. ■

Everything generalizes to the case of $T + 1$ periods and gradual release of information. With several periods, an asset can take many forms. For example, we could have *short-term assets* available for trade in a given period and having positive returns only in the next period. Or we could have *long-term assets* available at $t = 0$, tradeable at every period, and having positive returns only at the final period $t = T$. And, of course, there could be mixed cases of assets tradeable in a subset of periods and providing returns also in a subset of periods (not necessarily the same).

Again, the equivalence result of Proposition 19.D.1 generalizes if the asset structure is *complete*. Suppose, for example, that our asset structure is composed only of collections of short-term assets available and tradeable at any admissible date-event pair tE and paying contingent amounts of physical good 1 at the immediately succeeding date-event pairs. Denote by $S(tE)$ the number of successors at tE . If the number of assets available at tE is $K(tE)$ we can view the return matrix at tE as an $S(tE) \times K(tE)$ matrix $R(tE)$. The completeness condition is then the requirement that $\text{rank } R(tE) = S(tE)$ for all admissible date-events pairs tE . In

Section 19.D, the matrices $R(tE)$ were identity matrices, and so the asset structure there was complete. But, to repeat, the results of Section 19.D generalize to the complete, nondiagonal case.

A very interesting, and new, phenomenon is that if assets are long lived, and therefore repeatedly tradeable as information is gradually disclosed, it may be possible to implement the Arrow–Debreu equilibrium with much fewer than S assets. This is illustrated in Example 19.E.7.

Example 19.E.7: Suppose that $T = 2$ and the date–event tree unfolds as in Figure 19.E.2. In

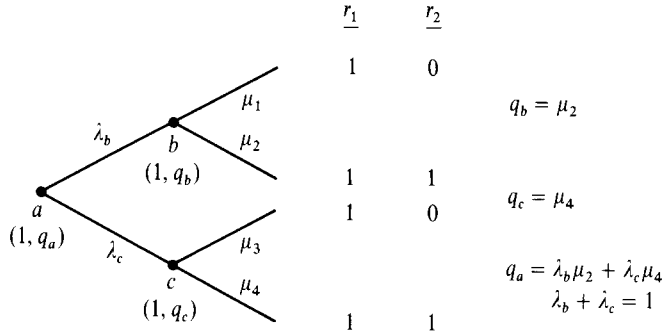


Figure 19.E.2
Construction of Arrow–Debreu prices from the equilibrium values of the asset prices of two sequentially traded assets.

the figure we have seven admissible date–events corresponding to the four terminal nodes, or states, to the initial node, denoted a , and to two intermediate nodes, denoted b and c . In particular, there are no more than two branches from any node. In this case we claim that, typically, two long-lived assets should suffice to guarantee that the Radner and the Arrow–Debreu equilibrium consumption allocations are the same.¹⁹ Suppose, to take a simple instance, that $L = 1$ and that our two assets have return vectors $r_1 = (1, 1, 1, 1)$ and $r_2 = (0, 1, 0, 1)$, payable at the terminal nodes. Consumption takes place only at the terminal nodes, but the assets may be traded both at the node a corresponding to $t = 0$ and at the nodes b and c corresponding to $t = 1$. We can normalize the price of the first asset (as well as the price of final consumption) to be 1 at every node.²⁰ Denote by q_a , q_b , and q_c the prices of the second asset at the respective nodes. By arbitrage (Proposition 19.E.1), applied at $t = 1$, there are $(\mu_1, \mu_2) \geq 0$ such that $\mu_1 + \mu_2 = 1$, $\mu_2 = q_b$, and $(\mu_3, \mu_4) \geq 0$ such that $\mu_3 + \mu_4 = 1$ and $\mu_4 = q_c$. Again by arbitrage (applied this time at $t = 0$), we must have $(\lambda_b, \lambda_c) \geq 0$ such that $\lambda_b + \lambda_c = 1$ and $q_a = \lambda_b q_b + \lambda_c q_c = \lambda_b \mu_2 + \lambda_c \mu_4$. This suggests considering the following Arrow–Debreu prices:

$$p = (\lambda_b \mu_1, \lambda_b \mu_2, \lambda_c \mu_3, \lambda_c \mu_4).$$

In Exercise 19.E.5 you are asked to show that, under the weak condition $q_b \neq q_c$, the set of final consumptions achievable through sequential trade with asset prices (q_a, q_b, q_c) is indeed the same as the set of final consumptions achievable with the four Arrow–Debreu contingent commodities at prices p .²¹ ■

19. We say “typically” because the existence of two assets is a necessary condition for completeness at every node but, strictly speaking, not a sufficient condition.

20. Note that, as should be the case, we do not normalize more than one price per budget constraint.

21. Incidentally, the assets prices provide a specific instance of what is called the *martingale property of asset prices*: at any node the price of an asset is the conditional expectation of the final returns, where the expectation is taken with respect to some probabilities, in our case the Arrow–Debreu prices.

19.F Incomplete Markets

In this section we explore the implications of having fewer than S assets, that is, of having an asset structure that is necessarily incomplete. We pursue this point in the two-period framework of the previous sections.²²

Markets may fail to exist for a number of reasons. One class of reasons refers to the informational asymmetries to be covered in Section 19.H: contracts for the delivery of goods can only be made contingent on states whose occurrence can be verified to the satisfaction of all contracting parties. Another class of reasons stems from transaction costs: the availability of a market is, after all, in the nature of a public good. Yet another variety of reasons comes from enforceability constraints: a promise to deliver one unit of good is worthless if delivery cannot be enforced.²³ This said, we shall not delve further into a theory of asset determination. We will rest content for the moment with taking the incomplete situation as a reasonable description of reality.

We begin by observing that when $K < S$ a Radner equilibrium need not be Pareto optimal. This is not surprising: if the possibilities of transferring wealth across states are limited, then there may well be a welfare loss due to the inability to diversify risks to the extent that would be convenient. Just consider the extreme case where there are no assets at all. Example 19.F.1 provides another interesting illustration of this type of failure.

Example 19.F.1: Sunspots. Suppose that preferences admit an expected utility representation and that the set of states S is such that, first, the probability estimates for the different states are the same across consumers (i.e., $\pi_{si} = \pi_{si'} = \pi_s$ for all i, i' , and s) and, second, that the states do not affect the fundamentals of the economy; that is, the Bernoulli utility functions and the endowments of every consumer i are uniform across states [i.e., $u_{si}(\cdot) = u_i(\cdot)$ and $\omega_{si} = \omega_i$ for all s]. Such a set of states is called a *sunspot* set. The question we shall address [first posed by Cass and Shell (1983)] is whether in these circumstances the Radner equilibrium allocations can assign varying consumptions across states. An equilibrium where this happens is called a *sunspot equilibrium*.²⁴

Under the assumption that consumers are strictly risk averse, so that the utility functions $u_i(\cdot)$ are strictly concave, any Pareto optimal allocation $(x_1, \dots, x_I) \in \mathbb{R}^{LSI}$ must be uniform across states (or state independent); that is, for every i we must have

22. For a general and advanced treatment, see Magill and Shafer (1991).

23. An example of a situation where enforceability would be helped is when we are dealing with the shares of a firm (the total endowments of this asset is, therefore, positive) and no short sales are possible. Enforceability is then guaranteed because the physical shares—the ownership claims to the firm—are actually transacted at $t = 0$. This sort of consideration, which, unfortunately, clashes with the very convenient assumption that unlimited short sales are possible, could explain why some assets may exist and others not: in order for the asset to exist, it helps if the random variable has a physical reality that can be exchanged at $t = 0$.

24. The term “sunspots” is old, but the current meaning is recent. In the XIX century, research on the “sunspot problem” tried to determine the influence on the fundamentals (e.g., on agriculture) that could make an unobservable signal (sunspots) have an effect on prices. The modern problem is to determine how an observable signal with no influence on fundamentals can nonetheless have, via expectations, an effect on prices.

$x_{1i} = x_{2i} = \dots = x_{si} = \dots = x_{Si}$. To see this, suppose that, for every i and s , we replace the consumption bundle of consumer i in state s , $x_{si} \in \mathbb{R}_+^L$, by the expected consumption bundle of this consumer: $\bar{x}_i = \sum_s \pi_s x_{si} \in \mathbb{R}_+^L$. The new allocation is state independent, and it is also feasible because

$$\sum_i \bar{x}_i = \sum_i \sum_s \pi_s x_{si} = \sum_s \pi_s \left(\sum_i x_{si} \right) \leq \sum_s \pi_s \left(\sum_i \omega_i \right) = \sum_i \omega_i.$$

By the concavity of $u_i(\cdot)$ it follows that no consumer is worse off:

$$\sum_s \pi_s u_i(\bar{x}_i) = u_i(\bar{x}_i) = u_i \left(\sum_s \pi_s x_{si} \right) \geq \sum_s \pi_s u_i(x_{si}) \quad \text{for every } i.$$

Because of the Pareto optimality of $(x_1, \dots, x_I)$, the above weak inequalities must in fact be equalities; that is, $u_i(\bar{x}_i) = \sum_s \pi_s u_i(x_{si})$ for every i . But, if so, then the strict concavity of $u_i(\cdot)$ yields $x_{si} = \bar{x}_i$ for every s . In summary: the Pareto optimal allocation $(x_1, \dots, x_I) \in \mathbb{R}^{LSI}$ is state independent.

From the state independence of Pareto optimal allocations and the first welfare theorem we reach the important conclusion that if a system of complete markets over the states S can be organized, then the equilibria are *sunspot free*, that is, consumption is uniform across states. In effect, traders wish to insure completely and have instruments to do so.

It turns out, however, that if there is not a complete set of insurance opportunities, then the above conclusion does not hold true. Sunspot-free, Pareto optimal equilibria always exist (just make the market “not pay attention” to the sunspot; see Exercise 19.F.1). But it is now possible for the consumption allocation of some Radner equilibria to depend on the state, and consequently to fail the Pareto optimality test. In such an equilibrium consumers expect different prices in different states, and their expectations end up being self-fulfilling. The simplest, and most trivial, example is when there are no assets whatsoever ($K = 0$). Then a system of spot prices $(p_1, \dots, p_S) \in \mathbb{R}^{LS}$ is a Radner equilibrium if and only if every p_s is a Walrasian equilibrium price vector for the spot economy defined by $\{(u_i(\cdot), \omega_i)\}_{i=1}^I$. If, as is perfectly possible, this economy admits several distinct Walrasian equilibria, then by selecting different equilibrium price vectors for different states, we obtain a sunspot equilibrium, and hence a Pareto inefficient Radner equilibrium. ■

We have seen that Radner equilibrium allocations need not be Pareto optimal, and so, in principle, there may exist reallocations of consumption that make all consumers at least as well off, and at least one consumer strictly better off. It is important to recognize, however, that this *need not* imply that a welfare authority who is “as constrained in interstate transfers as the market is” can achieve a Pareto optimum. An allocation that cannot be Pareto improved by such an authority is called a *constrained Pareto optimum*. A more significant and reasonable welfare question to ask is, therefore, whether Radner equilibrium allocations are constrained Pareto optimal. We now address this matter.²⁵

25. This is a typical instance of a *second-best* welfare issue. We have already encountered problems of this kind in Chapters 13 and 14, and we shall do so again in Chapter 22.

To proceed with the analysis we need a precise description of the constrained feasible set and of the corresponding notion of constrained Pareto optimality. This is most simply done in the context where there is a single commodity per state, that is, $L = 1$. The important implication of this assumption is that then the amount of consumption good that any consumer i gets in the different states is entirely determined by the portfolio z_i . Indeed, $x_{si} = \sum_k z_{ki} r_{sk} + \omega_{si}$. Hence, we can let

$$U_i^*(z_i) = U_i^*(z_{1i}, \dots, z_{Ki}) = U_i(\sum_k z_{ki} r_{1k} + \omega_{1i}, \dots, \sum_k z_{ki} r_{Sk} + \omega_{Si})$$

denote the utility induced by the portfolio z_i . The definition of constrained Pareto optimality is then quite natural.

Definition 19.F.1: The asset allocation $(z_1, \dots, z_I) \in \mathbb{R}^{KI}$ is constrained Pareto optimal if it is feasible (i.e., $\sum_i z_i \leq 0$) and if there is no other feasible asset allocation $(z'_1, \dots, z'_I) \in \mathbb{R}^{KI}$ such that

$$U_i^*(z'_1, \dots, z'_I) \geq U_i^*(z_1, \dots, z_I) \quad \text{for every } i,$$

with at least one inequality strict.

In this $L = 1$ context the utility maximization problem of consumer i becomes

$$\begin{aligned} \text{Max}_{z_i \in \mathbb{R}^K} \quad & U_i^*(z_{1i}, \dots, z_{Ki}) \\ \text{s.t.} \quad & q \cdot z_i \leq 0. \end{aligned}$$

Suppose that $z_i^* \in \mathbb{R}^K$ for $i = 1, \dots, I$, is a family of solutions to these individual problems, for the asset price vector $q \in \mathbb{R}^K$. Then $q \in \mathbb{R}^K$ is a Radner equilibrium price if and only if $\sum_i z_i^* \leq 0$.²⁶ Note that this has become now a perfectly conventional equilibrium problem with K commodities [see Exercise 19.F.2 for a discussion of the properties of $U_i^*(\cdot)$]. To it we can apply the first welfare theorem (Proposition 16.C.1) and reach the conclusion of Proposition 19.F.1.

Proposition 19.F.1: Suppose that there two periods and only one consumption good in the second period. Then any Radner equilibrium is *constrained Pareto optimal* in the sense that there is no possible redistribution of assets in the first period that leaves every consumer as well off and at least one consumer strictly better off.^{27, 28}

The situation considered in Proposition 19.F.1 is very particular in that once the initial asset portfolio of a consumer is determined, his overall consumption is fully determined: with only one consumption good, there are no possibilities for trade once the state occurs. In particular, second-period relative prices do not matter, simply because there are no such prices. Things change if there is more than one

26. Recall that, given z_i , the consumptions in every state are determined. Also, the price of consumption good in every state is formally fixed to be 1.

27. We reemphasize that the term *constrained* is appropriate. Wealth can be transferred across individuals and states only by trade in the given set of assets. To see how restrictive this can be, suppose that there are no assets. Then the welfare authority has no policy instrument whatsoever.

28. In our current discussion, all consumption takes place in the second period. This is a simplification that does not affect the validity of the proposition. If the welfare authority can also redistribute consumption that takes place in the first period, then the Radner equilibrium allocations will still be constrained Pareto optimal.

consumption good in the second period, or if there are more than two periods. Consider the two-period case with $L > 1$: Then we cannot summarize the individual decision problem by means of an indirect utility of the asset portfolio. The relative prices expected in the second period²⁹ also matter. This substantially complicates the formulation of a notion of constrained Pareto optimality. Be that as it may, there appears not to be a useful generalization of the “constrained Pareto optimal” concept in which we could assert the constrained Pareto optimality of Radner equilibrium allocations. Example 19.F.2, due to Hart (1975), makes the point. In it we have an economy with several Radner equilibria where two of them are Pareto ordered. That is, we have a Radner equilibrium that is Pareto dominated by another Radner equilibrium. To the extent that it seems natural to allow a welfare authority, at the very least, to select equilibria, it follows that the first equilibrium is not constrained Pareto optimal.³⁰

Example 19.F.2: Pareto Ordered Equilibria. Let $I = 2$, $L = 2$, and $S = 2$. There are no assets ($K = 0$). The two consumers have, as endowments, one unit of every good in every state. The utility functions are of the form $\pi_{1i}u_i(x_{11i}, x_{21i}) + \pi_{2i}u_i(x_{12i}, x_{22i})$. Note that although the probability assessments are different for the two consumers (these probabilities will be specified in a moment), the spot economies are identical in the two states. Suppose that this spot economy has several distinct equilibria (e.g., it could be the exchange economy in Figure 15.B.9). Let $p', p'' \in \mathbb{R}^2$ be the Walrasian prices for two of these equilibria and let $v_i(p)$ be the spot market utility associated with $u_i(\cdot, \cdot)$ and the spot price vector $p \in \mathbb{R}^2$. Suppose that $v_1(p') > v_1(p'')$. By Pareto optimality in the spot market, $v_2(p') < v_2(p'')$.

We now define two Radner equilibria. The first has equilibrium prices $(p_1, p_2) = (p', p'') \in \mathbb{R}^4$ and the second has $(p_1, p_2) = (p'', p') \in \mathbb{R}^4$. Because there is no possibility of transferring wealth across states, these are indeed Radner equilibrium prices and, moreover, they are so for any probability estimates π_i . However, the expected utility of these Radner equilibria for the different consumers depends on the π_i . We can see now that if consumer 1 believes that the first state is more likely than the second, that is, he has $\pi_{11} > \frac{1}{2}$, then he will prefer the first equilibrium to the second. Indeed, $\pi_{11} > \frac{1}{2}$ and $v_1(p') > v_1(p'')$ imply $\pi_{11}v_1(p') + \pi_{21}v_1(p'') > \pi_{11}v_1(p'') + \pi_{21}v_1(p')$. Similarly, if the second consumer believes that the second state is more likely than the first, that is, he has $\pi_{22} > \frac{1}{2}$, then he will *also* prefer the first equilibrium to the second: $\pi_{22} > \frac{1}{2}$ and $v_2(p') < v_2(p'')$ imply $\pi_{12}v_2(p') + \pi_{22}v_2(p'') > \pi_{12}v_2(p'') + \pi_{22}v_2(p')$. Thus, the Radner equilibrium with prices (p', p'') Pareto dominates the one with prices (p'', p') . ■

The consensus emerging in the literature seems to be that failures of restricted Pareto optimality (for natural meanings of this concept) are not only possible but even typical [Geanakoplos and Polemarchakis (1986)]. In Exercise 19.F.3 you are asked to develop a related optimality paradox: it is possible for the set of assets to expand and for everybody to be worse off at the new equilibrium! We shall not pursue the constrained optimality analysis

29. Or the relative prices of goods between the second and third period, if we are considering more than two dates.

30. That is, the first equilibrium is not Pareto optimal relative to any set of constrained feasible allocations that includes all Radner equilibrium allocations.

in any greater depth. At some point the analysis runs into the difficulty that it is hard to proceed sensibly without tackling the difficult problem of the determination of the asset structure.

We could also analyze the positive issues studied in Chapter 17 within an incomplete market setting. For existence, there is a new set of complexities related to the fact that unbounded short sales are possible. In some contexts this may lead to existence failures (see Exercise 19.F.4).³¹ New subtleties also arise for the issue of the determinacy of equilibria (i.e., the number and local uniqueness of equilibria). As we have seen in Section 17.D, with a complete asset structure we have generic finiteness. But with incomplete markets the nature of the assets (e.g., whether real or financial) matters, as may the size of S .

19.G Firm Behavior in General Equilibrium Models under Uncertainty

In the previous sections we have concentrated on the study of exchange economies. For once, this has not been just for simplicity. The consideration of production and firms is genuinely more difficult in a context of possibly incomplete markets. The reason relates to the issue of the objectives of the firm.³²

As before, we consider a setting with two periods, $t = 0$ and $t = 1$, and S possible states at $t = 1$. There are L physical commodities traded in the spot markets of period $t = 1$ and K assets traded at $t = 0$. There is no consumption at $t = 0$. The returns of the assets are in physical amounts of the good 1 (which we call the numeraire). The $S \times K$ return matrix is denoted R .

We introduce into our model a firm that produces a random amount of numeraire at date $t = 1$ (perhaps by means of inputs used at time $t = 0$, but we do not formalize this part explicitly). We let $(a_1, \dots, a_S)$ denote the state-contingent levels of production of the firm. There are also shares $\theta_i \geq 0$, with $\sum_i \theta_i = 1$, giving the proportion of the firm that belongs to consumer i . We take, for the rest of this section (except in the small-type paragraphs at the end) the natural point of view that the firm is an asset with return vector $a = (a_1, \dots, a_S)$ whose shares are tradeable in the financial markets at $t = 0$.³³

Suppose now that the firm can actually choose, within a range, its (random) production plan. Say, therefore, that there is a set $A \subset \mathbb{R}^S$ of possible choices of return

31. Unbounded short sales are at the origin of a discontinuity in the dependence on asset returns of the space of attainable wealth transfers across states. No matter how close asset returns (in dollar terms) may be to displaying a linear dependence, consumers can plan to attain, by using trades of very large magnitude, any wealth transfer in the subspace spanned by the asset returns. But when returns become exactly linearly dependent, this attainable subspace suddenly drops in dimension. As indicated, this can lead to an existence failure in some contexts. The model we have analyzed in this chapter is not, however, one of those. If, as here, in every state all assets have returns in a single good, which, moreover, is the same across assets, then the discontinuity does not arise.

32. The classic paper on this topic is Diamond (1967). For a more recent survey see Merton (1982).

33. A minor difference with the setting so far is that the firm does really produce the vector $(a_1, \dots, a_S)$, and therefore the total endowment of this asset is not zero. In fact, by putting $\sum_i \theta_i = 1$ we have normalized this total endowment to be 1.

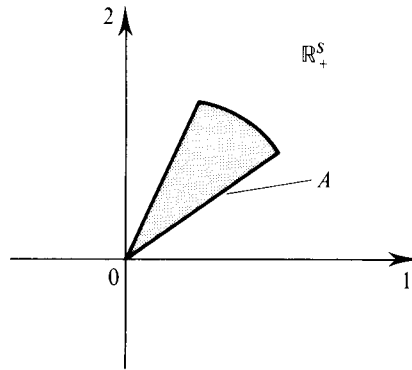


Figure 19.G.1
An example of
possible production
choices of the firm.

vectors $(a_1, \dots, a_S) \in A$ of the firm. See Figure 19.G.1 for the case where $S = 2$. We assume that the return vector $a \in A$ is chosen before the financial markets of period $t = 0$ open. Thus, the decision is made by the *initial* shareholders (since shares may be sold in period $t = 0$, the shareholders at the end of period $t = 0$ may be a different set). Which production plan should these initial owners choose? It turns out that the answer is very simple if A can be spanned by the existing assets and is very difficult if it cannot.

Definition 19.G.1: A set $A \subset \mathbb{R}^S$ of random variables is *spanned* by a given asset structure if every $a \in A$ is in the range of the return matrix R of the asset structure, that is, if every $a \in A$ can be expressed as a linear combination of the available asset returns.

If we assume, first, that A is spanned by R and, second, that we are dealing with a small project (i.e., all the possible productions $a \in A$ are small relative to the size of the economy; e.g., $a_s / \|\sum_i \omega_{si}\|$ is small for all s), then we are (almost) justified in taking the equilibrium spot prices $p = (p_1, \dots, p_S) \in \mathbb{R}^{L^S}$ and asset prices $q = (q_1, \dots, q_K) \in \mathbb{R}^K$ as constants independent of the particular production plan chosen by the firm.³⁴ For the asset prices $q \in \mathbb{R}^K$ the *market value* $v(a, q)$ of any production plan $a \in A$ can be computed by arbitrage: if $a = \sum_k \alpha_k r_k$ then $v(a, q) = \sum_k \alpha_k q_k$. In Exercise 19.G.1 you are asked to show that if the firm is added as a new asset to the given list of assets, and each production plan $a \in A$ is priced at its arbitrage value $v(a)$, then any budget-feasible consumption plan of any consumer can actually be reached without purchasing any shares of the firm (the fact can be deduced from Proposition 19.E.3). Thus, for fixed asset prices $q \in \mathbb{R}^K$ and spot prices $p = (p_1, \dots, p_S) \in \mathbb{R}^{L^S}$, the budget constraint of consumer i is³⁵

34. Both assumptions are important for this conclusion. Suppose for a moment that there are zero total endowments of the asset. Then, since the asset is redundant, Proposition 19.E.3 (see also Exercise 19.E.4) implies that at the Radner equilibrium the new asset is absorbed without any change in prices. What we are now assuming is that this remains approximately true if the total endowment of the asset is small (i.e., if the project is small).

35. Note that the value of the initial endowments at $t = 0$ is the value of the shares of the firm $\theta_i v(a, q)$.

$$B_{ai} = \{(x_{1i}, \dots, x_{Si}) \in \mathbb{R}_+^{LS} : \text{there is a portfolio } z_i \in \mathbb{R}^K \text{ such that} \\ p_s \cdot (x_{is} - \omega_{is}) \leq \sum_k p_{1s} r_{sk} z_{ki} \text{ for every } s, \text{ and } q \cdot z_i \leq \theta_i v(a, q)\}.$$

(19.G.1)

It follows from the form of this budget constraint that at constant prices every consumer-owner (i.e., any i with $\theta_i > 0$) faced with the choice between two production plans $a, a' \in A$, will prefer the one with higher market value. Indeed, if $v(a, q) \geq v(a', q)$ then $B_{a'i} \subset B_{ai}$. Thus, the objective of market value maximization will be the *unanimous* desire of the firm's initial owners.³⁶

If A is not spannable by the given asset structure we run into at least two serious difficulties. The first difficulty has to do with price *quoting* and is common to any commodity innovation problem. Without spanning, the value of a production plan $a \in A$ cannot be computed from current asset prices simply by arbitrage. The value is not, so to speak, implicitly quoted in the economy. Therefore, it would need to be anticipated by the agents of the economy from their understanding of the workings of the overall economy—no mean task.

The second difficulty, more specific to the financial context, has to do with price *taking*. Due to the possibility of unlimited short sales there is a discontinuity in the plausibility of the price-taking assumption. With spanning we can argue, as we did, that if the project is small then the effect of production decisions on asset prices, or on spot prices at $t = 1$, is also small. But if a new asset $a \in A$, no matter how small, is not generated by the current asset structure, then its availability increases the span of available wealth transfers by one whole extra dimension. The impact is therefore substantial, and may well have a dramatic effect on prices.³⁷ There is then no reason for owners' preferences over different production plans to be dictated merely by the increase in wealth at the prices prior to the introduction of the firm (see Exercise 19.G.2). These two difficulties, to repeat, are serious. There is no easy way out.

A variation of the above model entirely eliminates the asset role of the firm at $t = 0$. Let us assume that the firm's shares cannot be traded at $t = 0$.³⁸ If owners at $t = 0$ choose $a \in A$, this simply means that their endowments at $t = 1$ are modified by the random variable $\theta_i a$ that, recall, pays in good 1 (i.e., the new endowment of consumer i becomes $(\omega_{si} + (\theta_i a_s, 0, \dots, 0)) \in \mathbb{R}^L$ for every state s).

If $a \in A$ can be spanned, then we are as in the previous model. It does not matter whether shares of the firm can be sold or not at $t = 0$. In either case consumers can take positions in the asset markets that will guarantee that the resulting final consumptions at $t = 1$ are the same (Exercise 19.G.3).

If $a \in A$ cannot be spanned, matters are different. The good news is that, because no new tradeable asset is created at $t = 0$, the price-taking discontinuity problem disappears. The bad news is that there is now another difficulty: Because there is no market for the shares at $t = 0$, the value of the asset cannot be computed as a deterministic amount at $t = 0$. It is rather a

36. As long as the project is small and, consequently, the prices are almost constant.

37. Recall that short sales are possible. One way out of the dilemma is to put a bound on short sales of roughly the size of the possible production vectors. The cost of this assumption is that we have then to give up the theory of arbitrage pricing.

38. Or perhaps we are now at the end of period $t = 0$ and the financial markets have already closed.

random variable at $t = 1$ and therefore the risk attitudes of the consumer-owners will be essential to the determination of the preferred $a \in A$. In particular, unanimity of the consumer-owners should not be expected (see Exercise 19.G.4).

19.H Imperfect Information

Up to now we have concentrated the analysis on a model where spot trading for goods occurs under conditions of perfect information about the state of the world. In this section, we relax this feature by considering the possibility that this information is not perfect. In doing so we shall see that there is a key difference between the case of *symmetric* information (all traders have the same information), which is largely reducible to the previous theory, and the case of *asymmetric* information, where a host of new and difficult conceptual problems arise.

To focus on essentials, we deal with trade in a single period. You can think of it as the period $t = 1$ of the previous treatments. In this period, one of several states $s = 1, \dots, S$ can arise. Once a state occurs, we consider the simplest case in which there is a single spot market. In this market a first commodity (good, service, ...) is traded against a second good, to be thought of as money (thus, $L = 2$). The price of the second good is normalized to 1. We reserve the symbol $p \in \mathbb{R}$ for the price of the nonmonetary commodity.

There are I consumers. Given probabilities $\pi = (\pi_{1i}, \dots, \pi_{Si})$ over the states, a random consumption vector $x_i = (x_{1i}, \dots, x_{Si}) \in \mathbb{R}^{2S}$ is evaluated by consumer i according to an extended von Neumann-Morgenstern utility function:

$$U_i(x_i) = \sum_s \pi_{si} u_{si}(x_{si}),$$

where $u_{si}(\cdot)$ is consumer i 's Bernoulli utility function in state s . Consumer i also has an initial, state-dependent, endowment vector $\omega_i = (\omega_{1i}, \dots, \omega_{Si}) \in \mathbb{R}^{2S}$, and a *signal function* $\sigma_i(\cdot)$ assigning a real number $\sigma_i(s) \in \mathbb{R}$ to every state $s \in S$.

The state s occurs at the beginning of the period. We assume that, once this has happened, consumer i receives the initial endowment ω_{si} and the signal $\sigma_i(s) \in \mathbb{R}$. The interpretation is that consumer i can distinguish two states $s, s' \in S$ if and only if $\sigma_i(s) \neq \sigma_i(s')$.³⁹ Consistent with this interpretation, we require that the endowments be *measurable* with respect to the signal function, that is, $\omega_{si} = \omega_{s'i}$ whenever $\sigma_i(s) = \sigma_i(s')$ (thus, we can write ω_{si} as $\omega_{\sigma_i(s)i}$). In this manner the endowments of goods of consumer i do not reveal to him information about the state of the world that is not already revealed by the signal. After every consumer gets his signal, the spot market operates. Finally, at the end of the period, the state is revealed and consumption takes place.⁴⁰

39. Equivalently, as we did (in small type) in Section 19.B, we could use *information partitions* instead of explicit signal functions. The information partition $\mathcal{I}_i$ associated with the signal $\sigma_i(\cdot)$ is composed of the events $\{s \in S: \sigma_i(s) = c\}$ obtained by letting $c \in \mathbb{R}$ run over all possible values.

40. In a more general, multistage, situation, some information is revealed at the end of the period and the economy moves on to the next period.

Symmetric Information

We say that information is *symmetric* if any two states, $s, s' \in S$, are distinguishable by one consumer i if and only if they are distinguishable by every other consumer k ; that is, $\sigma_i(s) \neq \sigma_i(s')$ if and only if $\sigma_k(s) \neq \sigma_k(s')$. Thus, with symmetric information we can as well assume that all consumers share the same signal function. We therefore write $\sigma_i(\cdot) = \sigma(\cdot)$ for all i . We can think of $\sigma(\cdot)$ as a public signal.

With symmetric information the determination of the spot prices proceeds in a manner entirely parallel to what we have seen so far. Suppose that state s occurs. Then every consumer i receives the signal $\sigma(s)$ and initial endowments $\omega_{\sigma(s)i}$.⁴¹ From the signal and the prior probabilities $\pi_i = (\pi_{1i}, \dots, \pi_{Si})$, which we take to be strictly positive, he computes his posterior probabilities over the different states s' as

$$\pi_{s'i} | \sigma(s) = \frac{\pi_{s'i}}{\sum_{\{s'' : \sigma(s'') = \sigma(s)\}} \pi_{s''i}}$$

for any s' with $\sigma(s') = \sigma(s)$, and $\pi_{s'i} | \sigma(s) = 0$ otherwise. The utility of a consumption bundle $x_i \in \mathbb{R}^2$ conditional on the signal $\sigma(s)$ is then

$$u_i(x_i | \sigma(s)) = \sum_{s'} (\pi_{s'i} | \sigma(s)) u_{s'i}(x_i).$$

Therefore we have, conditional on s , a perfectly well-specified spot economy. Under the usual assumption of price-taking behavior, an equilibrium price will be generated. We write this price as $p(\sigma(s)) \in \mathbb{R}$.⁴²

The concept of an information signal function lends itself to interesting comparative statics exercises.

Definition 19.H.1: The signal function $\sigma' : S \rightarrow \mathbb{R}$ is *at least as informative* as $\sigma : S \rightarrow \mathbb{R}$ if $\sigma(s) \neq \sigma(s')$ implies $\sigma'(s) \neq \sigma'(s')$ for any pair s, s' . It is *more informative* if, in addition, $\sigma'(s) \neq \sigma'(s')$ for some pair s, s' with $\sigma(s) = \sigma(s')$.⁴³

Two arbitrary signal functions $\sigma(\cdot), \sigma'(\cdot)$ may well not be comparable by the “at least as informative” concept. If they are, it is natural to ask if the more informative signal leads to a welfare improvement. We pose the “improvement” question in an *ex ante* sense (see Exercise 19.H.1 for an *interim* and an *ex post* sense); that is, we want to compare the expected utility of the different consumers under $\sigma(\cdot)$ and under $\sigma'(\cdot)$ when the expectation is evaluated *before* s occurs.

41. The endowments could, of course, be the result of the execution of forward trades entered upon in the past. The measurability requirement, namely, the restriction that endowments depend on the state only through the signal, then captures the restriction that a forward contract can only be made contingent on information available at the time of the execution of the contract (strictly speaking, it should be conditional on information available at that time to the contract enforcing authority).

42. Because updated probabilities and utility functions depend only on the values of the signal, we have imposed the natural requirement that the clearing price depends also only on the signal; that is, we write $p(\sigma(s))$ rather than $p(s)$. Indeed, how could the (unmodeled) market mechanism manage to distinguish states that no consumer can distinguish?

43. In terms of the corresponding information partitions, the signal $\sigma'(\cdot)$ is more informative than $\sigma(\cdot)$ if the information partition of $\sigma'(\cdot)$ refines the information partition of $\sigma(\cdot)$.

Consider first the decision problem of a single consumer i . Suppose, for simplicity, that the spot price $p \in \mathbb{R}$ and the consumer wealth $w_i \in \mathbb{R}$ are given and are independent of s (a more general case is presented in Exercise 19.H.2). For any signal function $\sigma(\cdot)$ the consumer forms a consumption plan $x_i^{\sigma(\cdot)} \in \mathbb{R}^{2S}$ as follows: Subject to the restriction that $x_{si}^{\sigma(\cdot)} = x_{s'i}^{\sigma(\cdot)}$ whenever $\sigma(s') = \sigma(s)$, the consumer chooses, for every possible state s , a consumption $x_{si}^{\sigma(\cdot)}$ in his budget set that maximizes expected utility conditional on the signal $\sigma(s)$. The ex ante utility of the information signal function $\sigma(\cdot)$ is therefore $\sum_s \pi_{si} u_{si}(x_{si}^{\sigma(\cdot)})$.

Proposition 19.H.1: In the single-consumer problem, if the signal function $\sigma'(\cdot)$ is at least as informative as the signal function $\sigma(\cdot)$, then the ex ante utility derived from $\sigma'(\cdot)$, $\sum_s \pi_{si} u_{si}(x_{si}^{\sigma'(\cdot)})$, is at least as large as the ex ante utility derived from $\sigma(\cdot)$, $\sum_s \pi_{si} u_{si}(x_{si}^{\sigma(\cdot)})$.

Proof: The first observation is that for any $\sigma(\cdot)$, $x_i^{\sigma(\cdot)}$ solves the problem

$$\begin{aligned} \text{Max} \quad & \sum_s \pi_{si} u_{si}(x_{si}) \\ \text{s.t.} \quad & x_i \in B_i^{\sigma(\cdot)} = \{x_i \in \mathbb{R}^{2S} : px_{1si} + x_{2si} \leq w_i \text{ for every } s, \text{ and} \\ & x_{si} = x_{s'i} \text{ whenever } \sigma(s) = \sigma(s')\}. \end{aligned}$$

You are asked to verify this formally in Exercise 19.H.3.

The second observation is that if $\sigma'(\cdot)$ is at least as informative as $\sigma(\cdot)$, then $B_i^{\sigma(\cdot)} \subset B_i^{\sigma'(\cdot)}$. Again you should verify this in Exercise 19.H.3.

It follows that moving from $\sigma(\cdot)$ to $\sigma'(\cdot)$ we only expand the constraint set of the consumer's problem. In doing so, the maximum value cannot decrease. ■

The claim of Proposition 19.H.1 is intuitive: in an isolated single-person decision problem, a more informative signal does not upset the feasibility of any decision plan (hence, it brings about an expansion of the feasible set) because the decision maker always has the option not to act on the extra information provided by $\sigma'(\cdot)$ over $\sigma(\cdot)$. Unfortunately, this line of argument does *not* apply to a system with interacting decision makers. At the equilibrium the budget set of a single consumer may well be affected by the signal, even if the consumer does not use it. It suffices that the other consumers use it and that, as a result, the new information finds its way into the spot prices. Example 19.H.1 shows that, because of this, it is even possible for an increase of information to make *everybody* (ex ante) worse off.

Example 19.H.1: Suppose there are two consumers, two commodities, and two equally probable states $s = 1, 2$. In both states the two consumers' endowments of the two physical goods are $\omega_1 = (1, 0)$ for consumer 1 and $\omega_2 = (0, 1)$ for consumer 2. In total, therefore, there is one unit of every commodity in every state. The two consumers have the same von Neumann–Morgenstern expected utility function. Their state dependent Bernoulli utility function is

$$u_{si}(x_{1si}, x_{2si}) = \beta_s \sqrt{x_{1si}} + (1 - \beta_s) \sqrt{x_{2si}}$$

where $\beta_1 = 1$ and $\beta_2 = 0$. Thus, in state 1 the second good is worthless, while in state 2 the first good is worthless.

Suppose first that there is no information (i.e., there is no signal function distinguishing the two states). Then there is a single spot market where every

consumer chooses amounts (x_{1i}, x_{2i}) of the two commodities so as to maximize the expected utility function

$$\frac{1}{2}\sqrt{x_{1i}} + \frac{1}{2}\sqrt{x_{2i}}.$$

By symmetry (but nonetheless compute the first order conditions) we see that at equilibrium every consumer will get half of each commodity and have an expected utility of $1/\sqrt{2}$. Hence, in this no-information equilibrium every consumer has managed to insure against the possibility that the good he owns turns out to be worthless.

Suppose that instead we have a perfectly informative signal function revealing the state prior to the opening of the spot market. Then the spot market equilibrium will be different under the two states. What happens now is that in each state one good is known to be worthless and, therefore, there is no possibility of trade in the spot market: Each consumer consumes his endowment, receiving a utility of 1 in one state and of 0 in the other. Ex ante this means that under perfect information every consumer has an expected utility of $\frac{1}{2} < 1/\sqrt{2}$. Thus, we see that the availability of a more informative signal function makes everybody worse off. The reason is that the availability of information destroys insurance opportunities [a possibility first pointed out by Hirshleifer (1973)].⁴⁴ ■

Asymmetric Information

Suppose now that information is not symmetric; that is, the signal functions $\sigma_i(\cdot)$ are private and not necessarily the same across consumers. How to proceed then? A first thought is to proceed exactly as before. When s occurs every consumer observes $\sigma_i(s)$ and uses his signal function $\sigma_i(\cdot)$ to update probabilities and utility functions. This defines a spot economy to which we can associate in the usual way a spot market clearing price written as $p(\sigma_1(s), \dots, \sigma_I(s))$. Note that the price $p(\sigma_1, \dots, \sigma_I)$ depends on *all* the individual signals: One says that the price *aggregates* the information of the market participants. In particular, the *price function* $p(s) = p(\sigma_1(s), \dots, \sigma_I(s))$ need not be measurable with respect to the individual signal functions $\sigma_i(\cdot)$; that is, it may be that two states $s, s' \in S$ are *not* distinguishable by consumer i (i.e., $\sigma_i(s) = \sigma_i(s')$), but *are* distinguished by the market [i.e., $p(\sigma_1(s), \dots, \sigma_I(s)) \neq p(\sigma_1(s'), \dots, \sigma_I(s'))$]. This raises an important difficulty that we discuss by means of Example 19.H.2.

44. Suppose that our period is period 1 and that previous to it there is a period 0 in which forward trade could conceivably take place. Under the no-information scenario there can be no contingent trade at $t = 0$ since the two states cannot be distinguished at $t = 1$. The model considered, therefore, is as complete as it can be (hence, the equilibrium is Pareto optimal relative to the no-information structure). This is not so for the model with perfect information. There is then no informational impediment to the creation of a complete set of contingent markets at $t = 0$. With them the possibility of insurance would be restored. In general, if markets are complete [relative to the information signal function $\sigma(\cdot)$] then the equilibrium is a Pareto optimum [relative to $\sigma(\cdot)$] and, therefore, if information improves (and the corresponding additional markets are created) some traders may gain and some may lose (i.e., there may be distribution effects) but overall the new vector of ex ante expected utilities is at the frontier of an expanded utility possibility set. We can conclude, therefore, that if markets are always complete relative to the information signal (i.e., a forward market contingent on every signal takes place at $t = 0$), then not everyone can end up worse off if the information signal function improves.

Example 19.H.2: The economy has two goods and two consumers. The two consumers have identical utility functions $u_i(x_{1i}, x_{2i}) = \beta \ln x_{1i} + x_{2i}$. The parameter β is the same for the two consumers and it is uncertain. It can take the values $\beta = 1$ and $\beta = 2$ with equal probability. (Hence, we can think that there are two equally probable states: One yields $\beta = 1$ and the other $\beta = 2$.) The two consumers have deterministic endowments of one unit of the first good (because of quasilinearity with respect to the second good we do not need to specify endowments of the latter).

The first consumer has an informative signal $\sigma_1(\beta) = \beta$ that allows him to distinguish the two possible values of β . The second consumer is not informed: his signal function has $\sigma_2(\beta) = k$ for some constant k .

After nature has determined the value of β and the information $\sigma_1(\beta)$, $\sigma_2(\beta)$ has been transmitted to the two consumers, a spot market takes place (as usual, the price of the numeraire commodity is fixed to be 1). Since the first consumer knows β , his demand, given the price $p \in \mathbb{R}$ of the first good, is $x_{11}(p; \beta) = \beta/p$. The second, uninformed, consumer will equalize his expected marginal utility to the price. Hence, his demand function, which does not depend on β , is

$$x_{12}(p) = \frac{1}{p} \left[\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 2 \right] = \frac{3}{2p}. \quad (19.H.1)$$

Solving the market equilibrium equation $x_{11}(p; \beta) + x_{12}(p) = 2$ we get the equilibrium price function

$$p(\beta) = \frac{1}{4}(3 + 2\beta).$$

Note now that $p(1) \neq p(2)$. This means that the price reveals the informed consumer's information. If so, then it is logical to suppose that the uninformed consumer will try to use the observed market price to infer the unobserved value of β . There is really no good reason to prevent him from doing so. But, once he does, his demand will no longer be given by (19.H.1), and the price function $p(\beta)$ specified above will no longer clear markets for every possible value of β . This is the difficulty we wanted to illustrate. It suggests that what is needed is an equilibrium notion embodying a consistency requirement between the information revealed by prices and the information used by consumers. ■

We have argued in Example 19.H.2 that it makes sense to require that the information revealed by prices be taken into account by the consumers in making their consumption plans in the different spot markets. Suppose, therefore, that $p(s) = p(\sigma_1(s), \dots, \sigma_I(s))$ is an arbitrary price function. Interpret it as a specification of the prices expected to hold by the consumers at the different states. We could now view this price function as a public signal function and let any consumer use it in combination with his private signal. That is, when state s occurs, consumer i knows that the event $E_{p(s), \sigma_i(s)} = \{s' : p(s') = p(s) \text{ and } \sigma_i(s') = \sigma_i(s)\}$ has occurred and updates his probability estimates of any state $s' \in E_{p(s), \sigma_i(s)}$ to

$$\pi_{s'i} | p(s), \sigma_i(s) = \frac{\pi_{s'i}}{\sum_{\{s'' : s'' \in E_{p(s), \sigma_i(s)}\}} \pi_{s''i}}.$$

If, for the updated utility functions, the price $p(s)$ clears the spot market for every s , then we say that the price function $p(\cdot)$ is a *rational expectations equilibrium price function*.^{45, 46} This is expressed formally in Definition 19.H.2.

Definition 19.H.2: The price function $p(\cdot)$ is a *rational expectations equilibrium price function* if, for every s , $p(s)$ clears the spot market when every consumer i knows that $s \in E_{p(s), \sigma_i(s)}$ and, therefore, evaluates commodity bundles $x_i \in \mathbb{R}^2$ according to the updated utility function

$$\sum_{s'} (\pi_{s'i} | p(s), \sigma_i(s)) u_{s'i}(x_i).$$

We saw in Example 19.H.2 a situation in which all privately observed information is revealed by the spot market price. This suggests the following approach to the determination of a rational expectations equilibrium price function. Imagine (this is merely a hypothetical experiment) that all the individual signal functions are known to all consumers and that for every state the vector of signal values $(\sigma_1(s), \dots, \sigma_I(s))$ is made public and is, therefore, usable by all consumers to update probabilities and utilities. The market-clearing price function $\hat{p}(s) = \hat{p}(\sigma_1(s), \dots, \sigma_I(s))$ thus generated is called the *pooled information equilibrium price function*. If the values of $\hat{p}(\cdot)$ distinguish all possible values of $(\sigma_1, \dots, \sigma_I)$, that is, if $\hat{p}(s) \neq \hat{p}(s')$ whenever $\sigma_i(s) \neq \sigma_i(s')$ for some s, s' , and i , then we say the price function $\hat{p}(\cdot)$ is *fully revealing*. In other words, the price function is fully revealing if it distinguishes the occurrence of any two states that can be distinguished by some consumer.

We argue now that if the pooled information equilibrium price function $\hat{p}(\cdot)$ is fully revealing, then it must be a rational expectations equilibrium price function. For any s , $\hat{p}(s)$ is determined under the assumption that every i knows that $s \in \{s': \sigma_k(s') = \sigma_k(s) \text{ for all } k\}$. Because the pooled information equilibrium price function $\hat{p}(\cdot)$ is fully revealing, it follows that $\{s': \sigma_k(s') = \sigma_k(s) \text{ for all } k\} = \{s': \hat{p}(s') = \hat{p}(s)\}$. Hence, for any s , $\hat{p}(s)$ is a market-clearing price when every i knows that $s \in E_{\hat{p}(s), \sigma_i(s)}$. We conclude that $\hat{p}(\cdot)$ is a rational expectations equilibrium price function. In other words: If a pooled information equilibrium price function is fully revealing, then the pooled information used by consumers need not be obtained by violating any privacy constraint but can simply be derived from the public price signals.

Example 19.H.2 continued: If both consumers are fully informed, we have the demand functions $x_{11}(p) = x_{12}(p) = \beta/p$. Thus, in this case, the pooled information equilibrium price function is $\hat{p}(\beta) = \beta$. This pooled information equilibrium price function is fully revealing and therefore a rational expectations equilibrium. ■

45. For the concept of rational expectations equilibrium (including additional references), see Green (1973), Grossman (1977) and (1981), Lucas (1972), and Allen (1986).

46. In this section we concentrate on issues relating to information transmission more than on matters concerning spanning or completeness. But note that, as we pointed out in the case of symmetric information (in footnote 44), there is no conceptual difficulty in imagining that previous to the period $t = 1$ under consideration there has been contingent trade for the delivery at $t = 1$ of amounts of physical good conditional on the values of the public signals at $t = 1$ (we call the overall situation complete if such contingent markets exist for every possible value of the public signals). Observe that because the spot price constitutes a public signal, a possible instrument of contingent trade is an asset with returns conditional on the realized value of the spot market price at $t = 1$; options are instances of such assets.

In Example 19.H.3 the pooled information equilibrium price function is not fully revealing and fails to be a rational expectations equilibrium. In fact, in the example no rational expectations equilibrium price function exists.

Example 19.H.3: [Kreps (1977)] There are two commodities and two consumers with utility functions $u_1(x_{11}, x_{21}) = \beta \ln x_{11} + x_{21}$ and $u_2(x_{12}, x_{22}) = (3 - \beta) \ln x_{12} + x_{22}$. As in Example 19.H.2, there are two states yielding values $\beta = 1, 2$ with equal probability. Consumer 1 is completely informed [i.e., $\sigma_1(1) \neq \sigma_1(2)$] while consumer 2 is uninformed [i.e., $\sigma_2(\beta) = \text{constant}$]. In the two states there is a total endowment of the first good of 3 units.

Given a rational expectations equilibrium price function, which we can write as $p(\beta)$, we have two possibilities. Either $p(1) \neq p(2)$, so that the information is revealed and as a consequence $p(\beta)$ coincides with the pooled information equilibrium price function $\hat{p}(\beta)$, or $p(1) = p(2)$ so that the information is not revealed.

The first possibility cannot arise because, if the information is pooled, then for the values $\beta = 1$ and $\beta = 2$ the utility functions are the same in the two states (except that the utility functions of consumers 1 and 2 are switched) and so the spot equilibrium price is also the same in the two states. In fact, $p = 1$ is the price that clears the market in the two states.

But the second possibility cannot arise either. With a constant, nonrevealing, price function, the uninformed consumer has a demand that is independent of the state, whereas the demand of the first (informed) consumer is state dependent. Hence, the same price cannot clear the market in both states.

In summary: If we assume that information is transmitted at equilibrium, then in fact it is not. And if we assume that it is not transmitted, then it is. As a result, we can only conclude that no rational expectation equilibrium price function exists. ■

As we have seen, the concept of a fully revealing equilibrium provides a useful tool for the study of markets with asymmetric information. In applications it is more common that we encounter a slightly weaker and more natural version of the full revelation idea. In effect, in order for the pooled information equilibrium price function $\hat{p}(\cdot)$ to be a rational expectations equilibrium price function, we do not need that, for every s , $\hat{p}(\cdot)$ reveals precisely the vector of signals $(\sigma_1(s), \dots, \sigma_I(s))$; it is enough that it reveals a *sufficient statistic* for this vector [or a statistic that is sufficient for every consumer i in conjunction with the private signal $\sigma_i(\cdot)$]. More generally, all we need is that for every possible state s the expressed demand of every consumer i at the price $\hat{p}(s)$ is the same whether the consumer knows the pooled information signed functions $(\sigma_1(\cdot), \dots, \sigma_I(\cdot))$, and receives the signal vector $(\sigma_1(s), \dots, \sigma_I(s))$, or instead knows only the signal function $\hat{p}(\cdot)$ [or only $\hat{p}(\cdot)$ and $\sigma_i(\cdot)$].

Example 19.H.4: The basic economy is as in Example 19.H.2. Now, however, the signal of each consumer $i = 1, 2$ is $\sigma_i = \beta^2 + \varepsilon_i$. The ε_i , for $i = 1$ and 2 , are noise variables independently distributed and taking the values $\varepsilon_i = -2, -1, 0, 1, 2$ with equal probability.⁴⁷

47. All of this could be expressed in terms of underlying states $s = (\beta, \varepsilon_1, \varepsilon_2)$. We would need $2 \times 5 \times 5 = 50$ of them.

Suppose that information is pooled. Then:

- (i) If $\sigma_i = 4, 5$, or 6 for either $i = 1$ or $i = 2$, we know that $\beta = 2$ with probability 1 and therefore $\hat{p}(\sigma_1, \sigma_2) = \beta/2 = 1$.
- (ii) If $\sigma_i = -1, 0$, or 1 for either $i = 1$ or $i = 2$, we know that $\beta = 1$ with probability 1 and therefore $\hat{p}(\sigma_1, \sigma_2) = \beta/2 = \frac{1}{2}$.
- (iii) In the remaining cases, $\sigma_i = 2$ or 3 for both $i = 1$ and $i = 2$, the updated probabilities on the two values of β remain $\frac{1}{2}$: No useful information is transmitted. Hence the clearing price is $\hat{p}(\sigma_1, \sigma_2) = \frac{3}{2}$ (Exercise 19.H.4).

The price function $\hat{p}(\cdot)$ defined by (i) to (iii) is not fully revealing: Given the value of $\hat{p}(\cdot)$ we cannot deduce from it the specific values of σ_1 and σ_2 .⁴⁸ Yet, the price function $\hat{p}(\cdot)$ is sufficient to distinguish among cases (i) to (iii), and therefore the knowledge of the single function $\hat{p}(\cdot)$ can replace for every consumer the knowledge of the vector of functions $(\sigma_1(\cdot), \sigma_2(\cdot))$. We can say, therefore, that $\hat{p}(\cdot)$ is a sufficient statistic for the signals, and conclude that $\hat{p}(\cdot)$ is a rational expectations price function. ■

In Example 19.H.5 price function is not a sufficient statistic but it becomes so when combined with the private signal of any consumer.

Example 19.H.5: The basic economy and the signals are as in Example 19.H.4, but with three differences. First, there are I consumers. Second, the noise terms ε_i are now payoff relevant: in particular, $u_i(x_i) = (\beta + \varepsilon_i) \ln x_{1i} + x_{2i}$. Third, half the consumers have their noise uniformly distributed in the interval $[-\frac{2}{3}, \frac{2}{3}]$, whereas the other half is perfectly informed about β , that is, $\varepsilon_i = 0$.

The pooled information equilibrium price function is $\hat{p}(\beta, \varepsilon_1, \dots, \varepsilon_I) = \beta + (1/I)(\sum_i \varepsilon_i)$. Note that this price function reveals β [if $\beta = 1$ then $\hat{p}(\cdot) < 1.5$ with probability 1; if $\beta = 2$ then $\hat{p}(\cdot) > 1.5$ with probability 1] but not the individual values of ε_i . However, a consumer i that knows β and $\sigma_i = \beta^2 + \varepsilon_i$ also knows ε_i , and therefore, at any given price, expresses a demand that coincides with the pooled information demand. We conclude that the pooled information equilibrium is a rational expectation equilibrium. It is important to observe that, in contrast with Example 19.H.4, the equilibrium price function alone does not now provide a sufficient statistic. At the rational expectations equilibrium the individual utility maximization problems of half the consumers make essential use of the private signal functions. ■

Example 19.H.5 allows us to address another issue. Suppose that to our general model we add the feature that acquiring the information signal function $\sigma_i(\cdot)$ costs some small amount of money $\delta > 0$. Suppose also that I is large, so that, plausibly, the pooled information price function $\hat{p}(\cdot)$ is not very sensitive to any single consumer i failing to acquire his signal function $\sigma_i(\cdot)$. Then we have the following paradox [see Grossman and Stiglitz (1976)]: If the price function $\hat{p}(\cdot)$ is fully revealing (or is a sufficient statistic by itself), why will any consumer i pay δ for the signal function $\sigma_i(\cdot)$? Any one consumer would rather not do so and attempt to free ride on the information transmitted by the price system. But if everybody proceeds in this manner, then the price function cannot be fully revealing (there is nothing to reveal)! Example 19.H.5 suggests one way out of this paradox: it can be verified that in the example there is a sufficiently small $\delta > 0$, not dependent on the number I , such that at any fixed price p , a consumer i with nontrivial ε_i , even if he already knows β , has an incentive to pay δ for the improvement of information provided by his private signal (Exercise 19.H.5). ■

48. Recall that “fully revealing” does not mean that we get to know the value of β , only that we get to know the value of the signals.

Up to now, prices may have conveyed information about an exogenously occurring state. But in a world of asymmetric information, prices could also convey information on the consumers' endogenously chosen actions, and those actions could matter for individual utilities. For example, the final utility of a consumer may depend not only of the number of units consumed and on exogenous states but also on some other statistic depending on other consumers' actions. If we regard this statistic as a "state" then it is as if states were determined endogenously. To illustrate this point we consider another example—the market for used cars (also referred to as the "lemons market"). With this, we connect with the analysis of adverse selection in Section 13.B and bring this section to a close.

Example 19.H.6: The Lemons Market. Suppose that consumers fall into two types: potential buyers and potential sellers (of, say, used cars). There are many consumers and twice as many potential sellers as potential buyers. Potential sellers have one unit of the good and potential buyers buy either one unit or none. The peculiarity of this market is that commodities are of two kinds: good and bad. Half the potential sellers have a good product and half have a bad one. The quality, known to the potential sellers, is unrecognizable to the buyers at the moment of trade. A good commodity is worth 1 to the potential seller and 4 to the buyer. A bad commodity is worth nothing to every consumer.

We could call the *state of the market* the fraction $\alpha \in [0, 1]$ of the commodities supplied that are of good quality. If the state of the market is α , then a buyer paying p gets expected utility $4\alpha - p$. The problem is that the state of the market depends on the price (thus, as before, the price provides information about the utility derived from consuming one unit of the good). Indeed, for any $p > 0$, every unit of bad commodity will be supplied to the market. But for $p < 1$ no unit of the good commodity will be supplied, whereas for $p > 1$ every unit of the good commodity will be supplied. Therefore, we must have $\alpha = 0$ for $p < 1$, $\alpha \in [0, \frac{1}{2}]$ for $p = 1$, and $\alpha = \frac{1}{2}$ for $p > 1$. If a pair (α, p) satisfies these inequalities then we say that the pair (α, p) is admissible. Note that the inequalities can be equivalently expressed as $p \leq 1$ for $\alpha = 0$, $p = 1$ for $\alpha \in (0, \frac{1}{2})$ and $p \geq 1$ for $\alpha = \frac{1}{2}$.

A potential buyer will deduce $\alpha = 0$ if he observes $p < 1$, and $\alpha = \frac{1}{2}$ if $p > 1$. Potential buyers may or may not express a demand depending on this inference. It is natural to say that if at the admissible pair (α, p) the total demand is not larger than the total supply then we are at a *rational expectations* equilibrium. In fact, in our case any admissible pair (α, p) turns out to be a rational expectations equilibrium. Note, however, that for some (α, p) the supply is larger than the demand (e.g. at $\alpha = \frac{1}{2}$, $p = 3$ no demand is expressed).⁴⁹ ■

49. For simplicity we have chosen an example where it is always the case that the forthcoming demand is not larger than the forthcoming supply. It is because of this that all admissible pairs (α, p) may appear as equilibria. More generally, some of these pairs may be eliminated because forthcoming demand is larger than supply. It is also worth observing that it would not be legitimate to impose as an equilibrium condition that demand be larger than or equal to supply. Suppose, for example, that $\alpha = \frac{1}{2}$ and $p = 1.5$. Then total supply is 2 while total demand is 1. The usual argument (underlying the tâtonnement dynamics) for downward pressure on demand is that some frustrated seller would attempt to sell to some buyer at a price $1.5 - \varepsilon$. But, in the current context, how is the buyer to know that the commodity being offered is not of bad quality? Note that things would look different if it were the buyer who approached a random seller (this is the reason why the requirement that the demand be no larger than the supply is a natural equilibrium condition.) In contrast, with symmetric information it is of no consequence who approaches whom when a buyer and a seller meet. The lesson to be learned from this discussion is that with asymmetric information the particular disequilibrium story matters a lot. To push the analysis forward it is therefore appropriate to refer back to Chapter 13 where, in a more limited, partial equilibrium setting, we have already studied asymmetric information problems with the help of a methodology well suited to the consideration of this type of microstructure.

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EXERCISES

19.C.1^A There are S states. A consumer has, in every state s , a Bernoulli utility function $u_s(x_s)$, where $x_s \in \mathbb{R}_+^L$. Suppose that, for every s , $u_s(\cdot)$ is concave. Show that the expected utility function $U(x_1, \dots, x_S) = \sum_s \pi_s u_s(x_s)$ defined on $\mathbb{R}_+^{LS}$ is concave.

19.C.2^A For the model described in Example 19.C.2, show that the marginal rates of substitution along the Pareto set are as drawn in Figure 19.C.2; that is, at any point of the Pareto set the marginal rate of substitution is smaller than the ratio of probabilities.

19.C.3^A Consider an Arrow–Debreu equilibrium of the economy described in Section 19.C. Suppose that $L = 1$ and that preferences of every consumer i admit an expected utility representation with continuous, strictly concave, and strictly increasing Bernoulli utility functions (identical across states). For every state s denote by p_s , π_{si} , and x_{si} the equilibrium price for the s -contingent commodity, the subjective probability of consumer i for state s , and the consumption of consumer i in state s , respectively.

Denote by $\bar{p} = \sum_s p_s$ the price of uncontingent delivery of one unit of consumption.

Show that $\sum_s (\pi_{si} \bar{p} - p_s) x_{si} \geq 0$ for every i . [Hint: Use a revealed preference argument.] Interpret.

19.C.4^B There are a single consumption good, two states, and two consumers. Note that this allows the use of Edgeworth boxes.

Utility functions are of the expected utility form. Bernoulli utility functions are identical across states. That is,

$$U_1(x_{11}, x_{21}) = \pi_{11}u_1(x_{11}) + \pi_{21}u_1(x_{21})$$

and

$$U_2(x_{12}, x_{22}) = \pi_{12}u_2(x_{12}) + \pi_{22}u_2(x_{22}),$$

where x_{si} is the amount of s -contingent good consumed by consumer i and π_{si} is the subjective probability of consumer i for state s . We assume that every $u_i(\cdot)$ is strictly concave and differentiable.

The total initial endowments of the two contingent commodities are $\bar{\omega} = (\bar{\omega}_1, \bar{\omega}_2) \gg 0$. We assume that every consumer gets half of the random variable $\bar{\omega}$, that is, $(\omega_{11}, \omega_{21}) = \frac{1}{2}\bar{\omega}$ and $(\omega_{12}, \omega_{22}) = \frac{1}{2}\bar{\omega}$.

(a) Suppose that consumer 1 is risk neutral, consumer 2 is not, and both consumers have the same subjective probabilities. Show that at an interior Arrow–Debreu equilibrium consumer 2 insures completely.

(b) Suppose now that consumer 1 is risk neutral, consumer 2 is not, and the subjective probabilities of the two consumers are not the same. Show then that at an interior (Arrow–Debreu) equilibrium consumer 2 will not insure completely. Which is the direction of the bias in terms of the differences in subjective probabilities? Argue also that consumer 1 (the risk-neutral agent) will not gain from trade.

19.C.5^A Consider an economy such as the one introduced in Section 19.C but with only one commodity in each state. There is a number I of risk-averse consumers. Preferences admit an expected utility representation. Suppose that the Bernoulli utility functions of a consumer for the good are identical across states and that subjective probabilities are the same across individuals. Individual endowments vary from state to state. However, we assume that the total endowment is nonstochastic, that is, uniform across states (if, say, I is large and the realizations of individual endowments are identically and independently distributed, then the total endowment per capita is almost nonstochastic).

Set up the Arrow–Debreu trading problem. Show that the allocation in which every individual's consumption in every state is the average across states of his endowments is an equilibrium allocation.

19.D.1^A Consider the model of sequential trade of Section 19.D. The only difference is that we now assume that, for every s , the s -contingent commodity pays 1 dollar (rather than one unit of physical good 1) if state s occurs (and nothing otherwise). Write down the budget constraints corresponding to this model and discuss which price normalizations are possible.

19.D.2^A Show that in Example 19.D.1 the contingent trades of the two consumers are as claimed in the discussion of the example.

19.D.3^A Formulate a model similar to the two-period model of Section 19.D with the difference that consumption also takes place at period $t = 0$. Show that the result of Proposition 19.D.1 remains valid.

19.D.4^B Consider a three-period economy, $t = 0, 1, 2$, in which at $t = 0$ the economy splits into two branches and at $t = 1$ every branch splits again into two. There are H physical commodities and consumption can take place at the three dates.

(a) Describe the Arrow–Debreu equilibrium problem for this economy.

(b) Describe the Radner equilibrium problem. Suppose that at $t = 0$ and $t = 1$ there are contingent markets for the delivery of one unit of the first physical good at the following date.

(c) Argue that the conclusion of Proposition 19.D.1 remains valid.

19.E.1^B Consider an asset trading model such as that considered in Section 19.E. The only difference is that consumption is also possible at date $t = 0$. Assume for simplicity that the Bernoulli utility functions on consumption are state independent and additively separable across time; that is, $u_t(x_{0t}, x_{1t}) = u_{0t}(x_{0t}) + u_{1t}(x_{1t})$, where $x_{0t}, x_{1t} \in \mathbb{R}^L$.

(a) Argue along the lines of the second proof of Proposition 19.E.1 that the conclusion of Proposition 19.E.1 remains valid.

(b) Suppose now that there is a single physical good in every period. Express the multipliers μ_s in terms of marginal utilities of consumption.

19.E.2^A Show that if a primary asset with return vector $r \in \mathbb{R}^S$ separates states, that is, if $r_s \neq r_{s'}$ whenever $s \neq s'$, then it is possible to create a complete asset structure by using only options on this primary asset. You can assume that $r_s > 0$ for every s .

19.E.3^A Complete part (i) of the proof of Proposition 19.E.2 in the manner requested in the text.

19.E.4^B In text.

19.E.5^B Complete the verification that at the prices specified in Example 19.E.7 the set of consumptions achievable through sequential trade is the same as the set of consumptions achievable through ex ante trade of the four Arrow–Debreu commodities.

19.E.6^A There are two dates: At date 1 there are three states; at date 0 there is trade in assets. There are two basic assets whose return vectors in current dollars are

$$r_1 = (64, 16, 4) \quad \text{and} \quad r_2 = (0, 0, 1).$$

The market prices of these assets are $q_1 = 32$ and $q_2 = 1$, respectively. In the following you are asked to price by arbitrage a variety of derived assets.

(a) Suppose that one unit of a derived asset is described as “One unit of this asset confers the right to buy one unit of asset 1 at 75% of its spot value in period 1 (after the state of the world occurs).” Write the return vector of this asset and price it.

(b) The situation is the same as in (a) except that the asset is modified to read “One unit of this asset confers the right to buy one unit of asset 1 at 75% of its spot value in period 1 (after the state of the world occurs) provided the spot value is at least 10.”

(c) Suppose that the asset is as in (b) except that “at least 10” is replaced by “at least 19.” Write down the return vector and argue that this asset cannot be priced by arbitrage with the available primary assets.

(d) How would the analysis in (c) differ if we had in addition a riskless asset with a price equal to 1? (You do not need to compute the price explicitly.)

(e) Suppose that now the asset is further complicated to read “One unit of this asset confers, at the choosing of the holder, either 1 dollar in period 1 or the right to buy one unit of asset

1 at 75% of its spot value in period 1 (after the state of the world occurs).” Write the return vector of this asset and price it.

(f) The situation is the same as in (e) except that the asset is modified to read “One unit of this asset confers, at the choosing of the holder, either 1 dollar in period 1 or the right to buy one unit of asset 1 at 75% of its spot value in period 1 (after the state of the world occurs) provided this value is at least 10.”

19.F.1^B Consider the sunspot model of Example 19.F.1. Argue that, under the standard conditions on preferences, there is a sunspot-free equilibrium, whatever the asset structure.

19.F.2^A Consider (for the case $L = 1$) the utility function $U_i^*(\cdot)$ on asset portfolios defined in Section 19.F. Give sufficient conditions for $U_i^*(\cdot)$ to be continuous and concave. Show also that if returns are strictly positive then $U_i^*(\cdot)$ is strictly increasing.

19.F.3^C The aim of this exercise is to show that it is possible in an incomplete market situation for the number of assets to increase while at the same time *everybody* becomes worse off at the new Radner equilibrium. We do this in steps.

(a) Construct a two-consumer economy with two equally likely states and in which the distributional effects of trade are so biased that the sum of the utilities at the equilibrium with complete markets is smaller than the sum of the utilities at the equilibrium with incomplete markets.

(b) Now construct an economy that has four equally likely states and in which in the first two states the economy is as in (a), while in the other two states it is also as in (a) *except* that the roles of the two consumers are reversed.

(c) Show then that the asset structure in which there is a single asset allowing a transfer of wealth from the first two states to the other two yields an equilibrium that is better for every consumer than the equilibrium obtained if we add two new assets, one allowing a transfer of wealth between states 1 and 2 and the other doing the same between states 3 and 4.

19.F.4^C Exhibit an example in which with unlimited short sales a Radner equilibrium may not exist. This example requires returns denominated in more than one commodity. [*Hint*: Recall the no-arbitrage necessary condition.]

19.F.5^B Consider an economy with a single period, a single consumption good, and a single input (labor). All workers are identical *ex ante*. Each of them has a probability $\frac{1}{2}$ of being able to work (in which case the production is k units of output and work causes no disutility). With probability $\frac{1}{2}$, the worker is unable to work (is disabled). The utility of an amount c of consumption if able or disabled is $U_a(c)$, $U_d(c)$, respectively. Assume that the probability of disability is independent across workers and that there are sufficient workers that society operates on the expected value production possibility set.

(a) If there is a full set of Arrow–Debreu markets that are open before the disability is known, what is the equilibrium allocation of resources? (You may need to allow for the possibility of infinitely many states.)

(b) Assume now that it is impossible for others to observe whether an individual really is disabled or just claims to be so and stops working. Assume that insurance markets continue to exist as a competitive industry. Assume also that the condition “ $U'_a(c_a) = U'_d(c_d)$ implies $c_a > c_d$ ” is satisfied and that each individual purchases insurance only from a single company. Show that the competitive equilibrium (you should also define what this means) is the same as the one derived in (a).

(c) Continue to assume that it is impossible for others to observe whether a claimed disability is real. But assume now that $U'_a(c_a) = U'_d(c_d)$ implies $c_a < c_d$. Show then that the

equilibrium described in the answer to (b) is not reachable. Continuing to assume that each worker can purchase insurance from no more than one insurance company, determine the competitive equilibrium. Is it optimal relative to the allocations the government can achieve, supposing that the government also cannot observe disability?

19.G.1^A Justify expression (19.G.1). That is, suppose that every possible production plan of the firm can be spanned and that prices (in the asset and spot markets) are given. Now introduce the firm as a new asset. Show that any consumption plan of any consumer can be reached without purchasing any shares of the firm.

19.G.2^A Suppose that, in a two-consumer economy with $L = 1$, initially there is a single asset (which, therefore, goes untraded; recall that we do not allow consumption at $t = 0$). Now a firm is introduced that can produce the return vector $\varepsilon a \in \mathbb{R}_+^S$. The firm is owned, with equal shares, by the two consumers. Give an example in which, no matter how small ε may be (and letting the vector $a \in \mathbb{R}_+^S$ remain fixed), the introduction of the firm as an asset tradeable at $t = 0$ has the property that at the new equilibrium one consumer is significantly better off and the other is significantly worse off.

19.G.3^A Suppose that, in an economy with $L = 1$, a firm is introduced that can produce the single return vector $a \in \mathbb{R}^S$. The ownership share of consumer i is θ_i . Suppose also that the returns a can be spanned by existing assets. Show that the consumption bundles reachable at an equilibrium are the same under the following two scenarios. In the first the shares of the firm are traded at $t = 0$. In the second the asset is not traded and for every i the vector $\theta_i a$ is added to the initial endowments of consumer i .

19.G.4^A Suppose there are two consumers and $L = 1$. Initially there are no assets. We now introduce a firm with two possible return vectors $A = \{a^1, a^2\} \subset \mathbb{R}_+^S$. Ownership shares are the same for the two consumers. The firm's shares are not traded at $t = 0$. The production choice of the firm, a^1 or a^2 , is added to the endowments at $t = 1$ (half to each consumer). Show that it is possible for the two consumers not to be unanimous in their preference for a^1 or a^2 .

19.H.1^A We place ourselves in the framework of the single-consumer decision problem considered in Proposition 19.H.1.

(a) Show that if $\sigma(\cdot)$ is completely revealing [i.e., $\sigma(s) \neq \sigma(s')$ whenever $s \neq s'$] then $x_i^{\sigma(\cdot)}$ is ex post optimal in the sense that for every $s \in S$ the consumption allocation in state s is optimal for the spot economy that obtains at state s . Show also that this need not be true if $\sigma(\cdot)$ is not completely revealing.

(b) Show that $x_i^{\sigma(\cdot)}$ is interim optimal in the following sense: there is no allocation x_i measurable with respect to $\sigma(\cdot)$ and such that for some possible signal $\sigma(s)$ the expected utility of x_i [conditional on $\sigma(s)$] is larger than that corresponding to $x_i^{\sigma(\cdot)}$.

(c) Show that if $\sigma'(\cdot)$ is at least as informative as $\sigma(\cdot)$ then for every s the expected utility generated by $x_i^{\sigma'(\cdot)}$, conditional on $\sigma'(s)$, cannot be inferior to the expected utility generated by $x_i^{\sigma(\cdot)}$, conditional on $\sigma'(s)$.

(d) Show, similarly, that if $\sigma'(\cdot)$ is at least as informative as $\sigma(\cdot)$ then for every s the expected utility generated by $x_i^{\sigma'(\cdot)}$, conditional on $\sigma(s)$, cannot be inferior to the expected utility generated by $x_i^{\sigma(\cdot)}$, conditional on $\sigma(s)$.

19.H.2^A Argue that for the validity of Proposition 19.H.1 we may allow p and w_i to depend on the state. What is required is that, as functions, they be measurable with respect to the original $\sigma(\cdot)$ and that they remain unaltered as $\sigma(\cdot)$ is replaced by $\sigma'(\cdot)$.

19.H.3^A Complete the requested steps of the proof of Proposition 19.H.1.

19.H.4^A Complete the missing step in Example 19.H.4.

19.H.5^B Carry out the requested verification in Example 19.H.5.

19.H.6^C Consider the following two-consumer, two-commodity general equilibrium model. The (Bernoulli) utility functions of the two consumers are

$$u_1(x_{11}, x_{21}) = x_{11} + x_{21},$$

$$u_2(x_{12}, x_{22}) = (x_{12})^{1/2} + x_{22}.$$

Consumer 1's endowment of the second good is ω_{21} . He has no endowment of the first good. Consumer 2 has no endowments of the second good and his endowments of the first good depend on which of three equally likely states occurs. The respective levels in the three states are ω_{112} , ω_{122} , ω_{132} .

(a) Determine the Arrow–Debreu equilibrium of this economy. You can assume that the parameters are such that the equilibrium is interior.

(b) Suppose that the only possible markets are for the noncontingent delivery of the two goods. Set up the equilibrium problem. Is the equilibrium allocation a Pareto optimum?

(c) Suppose now that before any trade takes place, and before the endowments are revealed, the two consumers are told whether or not state 1 has occurred. After the revelation of this information (and before the values of the endowments are disclosed) non-contingent trade can take place. Set up the equilibrium problem as it depends on the information available.

(d) The setting is as in (c); the only difference is that contingent trade (after the revelation of the information on state 1) is now permitted.

(e) Compare ex ante (i.e., before any announcement is known) the expected utilities attained by the two consumers in the equilibria of (a), (b), (c), and (d); assume that all of these equilibria are interior. When can you assert that the information available in parts (c) and (d) is socially valuable?

19.H.7^A Suppose that there are two equally likely states. In every state there is a spot market where a consumption good (good 1) is exchanged against the numeraire, which we denote as good 2. There are two consumers. Their utilities are

	State 1	State 2
Consumer 1	$2 \ln x_{11} + x_{21}$	$4 \ln x_{11} + x_{21}$
Consumer 2	$4 \ln x_{12} + x_{22}$	$2 \ln x_{12} + x_{22}$

The total endowment of the first good equals 6 in the first state and $6 + \varepsilon$ in the second state. All the endowments of this good are received by the second consumer. Assume also that the endowments of numeraire for the two consumers are sufficient for us to neglect the possibility of boundary equilibria. The price of the numeraire is fixed to be one in the two states. The prices of the non-numeraire good in the two states are denoted (p_1, p_2) .

(a) Suppose that when uncertainty is resolved, both consumers know which state of the world has occurred. Determine the spot equilibrium prices $(\hat{p}_1(\varepsilon), \hat{p}_2(\varepsilon))$ in the two states (as a function of the parameter ε).

(b) Assume now that when a state occurs, consumer 2 knows the state but consumer 1 remains uninformed (i.e., he must keep thinking of the two states as equally likely). Determine, under this information setup (and assuming that prices cannot be used as signals), the spot equilibrium prices $(\bar{p}_1(\varepsilon), \bar{p}_2(\varepsilon))$ in the two states.

(c) The situation is as in (b), except that now we allow consumer 1 to deduce the state of the world from prices. That is, if $p_1 \neq p_2$ then consumer 1 is actually informed, but if $p_1 = p_2$ he is not informed. A pair of spot prices $(p_1^*(\varepsilon), p_2^*(\varepsilon))$ constitute rational expectations

equilibrium prices if they clear the two spot markets when consumer 1 derives information from $((p_1^*(\varepsilon), p_2^*(\varepsilon)))$ in the manner just described. Suppose that $\varepsilon \neq 0$ and derive a rational expectations equilibrium pair of prices.

(d) Show that if $\varepsilon = 0$ then there is no rational expectations equilibrium pair of prices. Compare with Example 19.H.3.

Equilibrium and Time

20.A Introduction

In this chapter, we present the basic elements of the extension of competitive equilibrium theory to an intertemporal setting. In the presentation, we try to maintain a balance between two possible approaches to the theory varying by the degree of emphasis on time.

A first approach contemplates equilibrium in time merely as the particular case of the general theory developed in the previous chapters in which commodities are indexed by time as one of the many characteristics defining them. This is a useful point of view (the display of the underlying unity of seemingly disparate phenomena is one of the prime roles of theory), and to a point we build on it. However, exclusive reliance on this approach would, in the limit, be self-defeating. It would reduce this chapter to a footnote to the preceding ones.

A second approach proceeds by stressing, rather than deemphasizing, the special structure of time. Again, we follow this line to some degree. Thus, every model discussed in this chapter accepts the open-ended infinity of time, or the fact that production takes time. Also, at the cost of some generality, we pursue our treatment under assumptions of stationarity and time separability that allow for a sharp presentation of the dynamic aspects of the theory.

Sections 20.B and 20.C are concerned with the description of, respectively, the consumption and the production sides of the economy.

Section 20.D is the heart of this chapter. It deals with the basic properties of equilibria (including definitions, existence, optimality, and computability) in the context of a single-consumer economy.

Section 20.E (which concentrates on steady states) and Section 20.F (which is general) study the dynamics of the single-consumer model.

Section 20.G considers economies with several consumers. The message of this section is that, as long as the Pareto optimality of equilibrium is guaranteed, the qualitative aspects of the positive theory of Chapter 17 extend to the more general situation and, moreover, that the properties of individual equilibria identified by the single-consumer methodology remain valid in the broader context.

Section 20.H gives an extremely succinct account of overlapping-generations

economies, a model of central importance in modern macroeconomic theory. Our interest in it is twofold: on the one hand, we want to display it as yet another instance of a useful equilibrium model; on the other hand, we want to point out that it is an example that, because of the infinity of generations, does not fit the general model of Section 20.G, and one that gives rise to some new and interesting issues having to do with the optimality and the multiplicity of equilibria.

Section 20.I gathers some remarks on nonequilibrium considerations (short-run equilibrium and tâtonnement stability, learning, and so on).

For pedagogical purposes, the entire chapter deals only with the deterministic version of the theory. The unfolding of time is a line, not a tree. A full synthesis of the approaches of Chapter 19 (on uncertainty) and the current one (on time) is possible. However, we view its presentation as advanced material beyond the scope of this textbook. The account of Stokey and Lucas with Prescott (1989) constitutes an excellent introduction to the general theory.

A point of notation: in this chapter $\sum_t$ always means $\sum_{t=0}^{\infty}$, that is, $\lim_{T \rightarrow \infty} \sum_{t=0}^T$. When the sum does not run from $t = 0$ to $t = \infty$ the two end-points of the sum are explicitly indicated.

20.B Intertemporal Utility

In this chapter, we assume that there are infinitely many dates $t = 0, 1, \dots$, and that the objects of choice for consumers are *consumption streams* $c = (c_0, \dots, c_t, \dots)$ where $c_t \in \mathbb{R}_+^L$, $c_t \geq 0$.¹ To keep things simple, we will consider only consumption streams that are *bounded*, that is, that have $\sup_t \|c_t\| < \infty$.

Rather than proceed from the most general form of preferences over consumption streams to the more specific, we instead introduce first the very special form that we assume throughout this chapter (except for Sections 20.H and 20.I); we subsequently discuss its special properties from a general point of view.

It is customary in intertemporal economies to assume that preferences over consumption streams $c = (c_0, \dots, c_t, \dots)$ can be represented by a utility function $V(c)$ having the special form

$$V(c) = \sum_{t=0}^{\infty} \delta^t u(c_t) \quad (20.B.1)$$

where $\delta < 1$ is a *discount* factor and $u(\cdot)$, which is defined on $\mathbb{R}_+^L$, is strictly increasing and concave. This chapter will be no exception to this rule: Throughout it we assume that preferences over consumption streams take this form. However, we comment here, at some length, on six aspects of this utility function. As a matter of notation, given a consumption stream $c = (c_0, \dots, c_t, \dots)$, we let $c^T = (c_0^T, c_1^T, \dots)$ denote the T -period “backward shift” consumption stream, that is, the stream $(c_0^T, c_1^T, \dots)$ with $c_t^T = c_{t+T}$ for all $t \geq 0$.

(1) *Time impatience*. The requirement that future utility is discounted (i.e., that $\delta < 1$), implies *time impatience*. That is, if $c = (c_0, c_1, \dots, c_t, \dots)$ is a non-zero consumption stream, then the (forward-) shifted consumption stream $c' = (0, c_0, c_1, \dots, c_{t-1}, \dots)$ is strictly worse than c (see Exercise 20.B.1). It is an

1. We use the terms “stream,” “trajectory,” “program,” and “path” synonymously.

assumption that is very helpful in guaranteeing that a bounded consumption stream has a finite utility value [i.e., guarantees that the sum in (20.B.1) converges], thus allowing us to compare any two such consumption streams² and making possible the application of the machinery of the calculus. There is a strand of opinion that views this technical convenience as the real reason for the fundamental role that the assumption of time discounting plays in economics. This skeptical view on the existence of substantive reasons³ is excessive. An implication of time discounting is that the distant future does not matter much for current decisions, and this feature seems more realistic than its opposite.

A possible interpretation, and defense, of the discount factor δ views it as a probability of survival to the next period. Then $V(c)$ is the expected value of lifetime utility. For another interpretation, see (6) below.

(2) *Stationarity*. A more general form of the utility function would be

$$V(c) = \sum_{t=0}^{\infty} u_t(c_t). \quad (20.B.2)$$

The form (20.B.1) corresponds to the special case of (20.B.2) in which $u_t(c_t) = \delta^t u(c_t)$. This special form can be characterized in terms of *stationarity*. Consider two consumption streams $c \neq c'$ such that $c_t = c'_t$ for $t \leq T-1$: that is, the two streams c and c' are one and the same up to period $T-1$ and differ only after $T-1$. Observe that the problem of choosing at $t = T$ between the current and future consumptions in c and c' is the same problem that a consumer would face at $t = 0$ in choosing between the consumption streams c^T and c'^T , the T backward shifts of c and c' , respectively. Then *stationarity* requires that

$$V(c) \geq V(c') \quad \text{if and only if} \quad V(c^T) \geq V(c'^T).$$

It is a good exercise to verify that (20.B.1) satisfies the stationarity property and that the property can be violated by utility functions of the form $V(c) = \sum_t \delta_t^t u(c_t)$, that is, with a time-dependent discount factor (Exercise 20.B.2).

The property of stationarity should *not* be confused with the statement asserting that if the consumption streams c and c' coincide in the first $T-1$ periods and a consumer chooses one of these streams at $t = 0$, then she will not change her mind at T . This “property” is tautologically true: at both dates we are comparing $V(c)$ and $V(c')$.⁴ The stationarity experiment compares $V(c)$ and $V(c')$ at $t = 0$, but at period T it compares the utility values of the future streams shifted to $t = 0$, that is, $V(c^T)$ and $V(c'^T)$. Thus, stationarity says that in the context of the form (20.B.2), the preferences over the future are independent of the *age* of the decision maker.

Time stationarity is not essential to the analysis of this chapter (except for Sections 20.E and 20.F on dynamics), but it saves substantially on the use of subindices.

2. Hence, the completeness of the preference relation on consumption streams is guaranteed.

3. Ramsey (1928) called the assumption a “weakness of the imagination.”

4. This property is often called *time consistency*. Time inconsistency is possible if tastes change through time (recall the example of Ulysses and the Sirens in Section 1.B!), but, as we have just argued, it must necessarily hold if the preference ordering over consumption streams $(c_0, \dots, c_t, \dots)$ does not change as time passes. In line with the entire treatment of Part IV, we maintain the assumption of unchanging tastes throughout the chapter.

(3) *Additive separability.* Two implications of the additive form of the utility function are that at any date T we have, first, that the induced ordering on consumption streams that begin at $T + 1$ is independent of the consumption stream followed from 0 to T , and, second, that the ordering on consumption streams from 0 to T is independent of whatever (fixed) consumption expectation we may have from $T + 1$ onward (see Exercise 20.B.3). In turn, these two separability properties imply additivity; that is, if the preference ordering over consumption streams satisfies these separability properties, then it can be represented by a utility function of the form $V(c) = \sum_t u_t(c_t)$ [this is not easy to prove, see Blackorby, Primont and Russell (1978)].

How restrictive is the assumption of additive separability? We can make two arguments in its favor: the first is technical convenience; the second is a vague sense that what happens far in the future or in the past should be irrelevant to the relative welfare appreciation of current consumption alternatives. Against it we have obvious counter-examples: Past consumption creates habits and addictions, the appreciation of a particularly wonderful dish may depend on how many times it has been consumed in the last week, and so on. There is, however, a very natural way to accommodate these phenomena within an additively separable framework. We could, for example, allow for the form $V(c) = \sum_t u_t(c_{t-1}, c_t)$. Here the utility at period t depends not only on consumption at date t but also on consumption at date $t - 1$ (or, more generally, on consumption at several past dates). We can formulate this in a slightly different way. Define a vector z_t of “habit” variables and a *household production technology* that uses an input vector c_{t-1} at $t - 1$ to jointly produce an output vector c_{t-1} of consumption goods at $t - 1$ and a vector $z_t = c_{t-1}$ of “habit” variables at t . Then, formally, u_t depends only on time t variables and total utility is $\sum_t u_t(z_t, c_t)$. In summary: additive separability is less restrictive than it appears if we allow for household production and a suitable number (typically larger than 1) of current variables.

(4) *Length of period.* The plausibility of the separability assumption, which makes the enjoyment of current consumption independent of the consumption in other periods, depends on the length of the period. Because even the most perishable consumption goods have elements of durability in them (in the form, for example, of a flow of “services” after the act of consumption), the assumption is quite strained if the length of the elementary period is very short. What determines the length of the period? To the extent that our model is geared to competitive theory, this period is institutionally determined: it should be an interval of time for which prices can be taken as constant. On a related point, note that the value of δ also depends, implicitly, on the length of the period. The shorter the period, the closer δ should be to 1.

(5) *Recursive utility.* With the form (20.B.1) for the utility function, we have $V(c) = u(c_0) + \delta V(c^1)$ for any consumption stream $c = (c_0, c_1, \dots, c_t, \dots)$. If we think of $u = u(c_0)$ as current utility and of $V = V(c^1)$ as future utility, we see that the marginal rate of substitution of current for future utility equals δ and is therefore independent of the levels of current and future utility. The *recursive utility model* [due to Koopmans (1960)] is a useful generalization of (20.B.1) that combines two features: it allows this rate to be variable but, as in the additively separable case, it has the property that the ordering of future consumption streams is independent of the consumption stream followed in the past.

The recursive model goes as follows. Denote current utility by $u \geq 0$ and future utility by $V \geq 0$. Then we are given a current utility function $u(c_t)$ and an *aggregator* function $G(u, V)$ that combines current and future utility into overall utility. For example, in the separable additive case we have $G(u, V) = u + \delta V$. More generally we could also have, for example, $G(u, V) = u^\alpha + \delta V^\alpha$, $0 < \alpha \leq 1$. In this case, the indifference curves in the (u, V) plane are not straight lines. The utility of a consumption stream $c = (c_0, \dots, c_t, \dots)$ could then be computed recursively from

$$V(c) = G(u(c_0), V(c^1)) = G(u(c_0), G(u(c_1), V(c^2))) = \dots \quad (20.B.3)$$

For (20.B.3) to make sense we must be able to argue that the influence of $V(c^T)$ on $V(c)$ will become negligible as $T \rightarrow \infty$ [so that $V(c)$ can be approximately determined by taking a large T and letting $V(c^T)$ have an arbitrary value]. This amounts to an assumption of time impatience. In applications, it will typically not be necessary to compute $V(c)$ explicitly. See Exercise 20.B.4 for more on recursive utility.

(6) *Altruism*. The expression $V(c) = u(c_0) + \delta V(c^1)$ suggests a multigeneration interpretation of the single-consumer problem (20.B.1). Indeed, if generations live a single period and we think of generation 0 as enjoying her consumption according to $u(c_0)$, but caring also about the utility $V(c^1)$ of the next generation according to $\delta V(c^1)$, then $V(c) = u(c_0) + \delta V(c^1)$ is her overall utility. If every generation is similarly altruistic, then we conclude, by recursive substitution, that the objective function of generation 0 is precisely (20.B.1). The entire “dynasty” behaves as a single individual. With this we also have another justification for $\delta < 1$. The inequality means then that the members of the current generation care for their children, but not quite as much as for themselves. See Barro (1989) for more on these points.

20.C Intertemporal Production and Efficiency

Assume that there is an infinite sequence of dates $t = 0, 1, \dots$. In each period t , there are L commodities. If it facilitates reading, you can take $L = 2$ and interpret the commodities as labor services and a generalized consumption–investment good (see Example 20.C.1). One of the great advantages of vector notation, however, is that in some cases—and this is one—there is no novelty involved in the general case. Thus, while you think you are understanding the simple problem, you are at the same time understanding the most general one.

We shall adopt the convention that goods are *nondurables*. This is a convention because, in order to make a good durable, it suffices to specify a storage technology whose role is, so to speak, to transport the commodity through time.

If we were exogenously endowed with some amount of resources (e.g., some initial capital and some amount of labor every period), we would ask what we could do with them. To give an answer, we need to specify the *production technology*. We already know from Chapter 5 how to do this formally by means of the concept of a *production set* (or a production transformation function, or a production function). With minimal loss of generality, we will restrict our technologies to be of the following form: the production possibilities at time t are entirely determined by the production decisions at the most recent past, that is, at time $t - 1$. If we keep in mind that we can always define new intermediate goods (such as different vintages of a machine),

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$$V(c) = G(u(c_0), V(c^1)) = G(u(c_0), G(u(c_1), V(c^2))) = \dots \quad (20.B.3)$$

For (20.B.3) to make sense we must be able to argue that the influence of $V(c^T)$ on $V(c)$ will become negligible as $T \rightarrow \infty$ [so that $V(c)$ can be approximately determined by taking a large T and letting $V(c^T)$ have an arbitrary value]. This amounts to an assumption of time impatience. In applications, it will typically not be necessary to compute $V(c)$ explicitly. See Exercise 20.B.4 for more on recursive utility.

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If we were exogenously endowed with some amount of resources (e.g., some initial capital and some amount of labor every period), we would ask what we could do with them. To give an answer, we need to specify the *production technology*. We already know from Chapter 5 how to do this formally by means of the concept of a *production set* (or a production transformation function, or a production function). With minimal loss of generality, we will restrict our technologies to be of the following form: the production possibilities at time t are entirely determined by the production decisions at the most recent past, that is, at time $t - 1$. If we keep in mind that we can always define new intermediate goods (such as different vintages of a machine),

and also that we can always define periods to be very long, we see that the restriction is minor.

Thus, the technological possibilities at t will be formally specified by a production set $Y \subset \mathbb{R}^{2L}$ whose generic entries, or *production plans*, are written $y = (y_b, y_a)$. The indices b and a are mnemonic for “before” and “after.” The interpretation is that the production plans in Y cover two periods (the “initial” and the “last” period) with $y_b \in \mathbb{R}^L$ and $y_a \in \mathbb{R}^L$ being, respectively, the production plans for the initial and the last periods. As usual, negative entries represent inputs and positive entries represent outputs.

We impose some assumptions on Y that are familiar from Section 5.B:

- (i) Y is *closed and convex*.
- (ii) $Y \cap \mathbb{R}_+^{2L} = \{0\}$ (*no free lunch*).
- (iii) $Y - \mathbb{R}_+^{2L} \subset Y$ (*free disposal*).

An assumption specific to the temporal setting is the requirement that inputs not be used *later* than outputs are produced (i.e., production takes time). This is captured by

- (iv) If $y = (y_b, y_a) \in Y$ then $(y_b, 0) \in Y$ (*possibility of truncation*).

In words, (iv) says that, whatever the production plans for the initial period, not producing in the last period is a possibility. A simple case is when $y_{at} \geq 0$ for every $y \in Y$, that is, when all inputs are used in the initial period. Then (iv) is implied by the free-disposal property (iii).

Example 20.C.1: Ramsey–Solow Model.⁵ Assume that there are only two commodities: A consumption–investment good and labor. It will be convenient to describe the technology by a production function $F(k, l)$. To any amounts of capital investment $k \geq 0$ and of labor input $l \geq 0$, applied in the initial period, the production function assigns the *total* amount $F(k, l)$ of consumption–investment good available at the last period. Then

$$Y = \{(-k, -l, x, 0) : k \geq 0, l \geq 0, x \leq F(k, l)\} - \mathbb{R}_+^4.$$

Note that labor is a primary factor; that is, it cannot be produced. ■

Example 20.C.2: Cost-of-Adjustment Model. Suppose that there are three goods: capacity, a consumption good, and labor. With the amounts k and l of invested capacity and labor at the initial period, one gets $F(k, l)$ units of consumption good output at the last period. This output can be transformed into invested capacity at the last period at a cost of $k' + \gamma(k' - k)$ units of consumption good for k' units of capacity, where $\gamma(\cdot)$ is a convex function satisfying $\gamma(k' - k) = 0$ for $k' < k$ and $\gamma(k' - k) > 0$ for $k' > k$. The term $\gamma(k' - k)$ represents the cost of adjusting capacity upward in a given period relative to the previous period. (Note the marginal cost of doing so increases with invested capacity of the period.) Formally, the production set Y is

$$Y = \{(-k, 0, -l, k', x, 0) : k \geq 0, l \geq 0, k' \geq 0, x \leq F(k, l) - k' - \gamma(k' - k)\} - \mathbb{R}_+^6. \quad \blacksquare$$

5. See Ramsey (1928) and Solow (1956). The same model was also introduced in Swan (1956).